GRIFFIN





AIAA DBF 2025-2026 DESIGN REPORT

THE UNIVERSITY OF HONG KONG

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Abbreviations, Acronyms and Symbols

A	Amps
AIAA	American Institute of Aerospace and Aeronautics
AC	Aerodynamic Center
AoA	Angle of Attack
AR	Aspect Ratio
BEC	Battery Eliminated Circuit
b	Wingspan
C_D	Coefficient of Drag
$C_{D,0}$	Coefficient of zero-lift drag
C_L	Coefficient of Lift
CAD	Computer-Aided-Design
CFD	Computational-Fluid-Dynamics
CFRP	Carbon Fibre Reinforced Polymer
CG	Center of Gravity
CLK-Y	Clark-Y
COTS	Commercial Off-the Shelf
CR	Cargo
ρ	Density of air
DBF	Design Build and Fly
D	Drag force
ζ	Damping Ratio
E	Total Energy Consumption
η	Efficiency
ESC	Electronic Speed Controller
ft	Imperial Foot
FEA	Finite Element Analysis
f	Frequency
F_1	Natural Frequency
g	Gravitational constant
GM	Ground Mission
GPS	Global Positioning System
HKU	The University of Hong Kong
in	inches
I	Current
K	Oswald Efficiency
L	Lift force
LiPo	Lithium-Polymer
lb	Imperical Pounds
lbf	Pounds-force
m	meter
M#	Mission Number
MAC	Average chord length
MTOW	Maximum Take-off Weight
mAh	Mili-ampere per hour
M1	Mission 1
M2	Mission 2
M3	Mission 3
oz	ounce
τ	Period
P	Battery Power
PS	Passengers
RAC	Rated Airplane Cost
RPM	Revolution per Minute
S	Planform Area
STOL	Short Take-Off and Landing
s	Second
T/W	Thrust to weight ratio
t	Time required
t_2	Time to half the Amplitude
V	Voltage
V_{cruise}	Cruise Velocity
UBEC	Universal Battery Eliminated Circuit
W	Watts
WFX	Wortmann-FX 63-137
Wh	Watt-hours

1 Executive Summary

This design report details the aerodynamic and structural design, manufacturing and testing of GRIFFIN, the bush model aircraft from the University of Hong Kong DBF team for the 2026 American Institute of Aeronautics and Astronautics (AIAA) Design, Build, Fly (DBF) competition. The team designed an aircraft with 4.1-foot wingspan aircraft carries a payload consisting of 3 passengers and 1 cargo and a 13-foot banner to complete three flight missions and one ground mission. Mission 1 (M1) is a delivery flight that demonstrates the aircraft's flight capabilities without a payload. Mission 2 (M2) is a carriage flight that includes passengers and cargo. Mission 3 (M3) is a launch flight that demonstrates the stowing, towing, deployment and release of banner. Finally, Ground Mission (GM) is a timed demonstration of aircraft assembly, payload installation, banner mechanism installation, and flight control checks.

The design of the aircraft was divided into three phases: conceptual, preliminary, and detailed design. During the conceptual design phase, the competition mission requirements were analyzed and mapped into design constraints to determine the overall aircraft configuration, together with geographical considerations which turns Griffin into a portable, detachable aircraft to be shipped to US. The preliminary design phase included trade studies focused on aircraft sizing, performance, and stability. The detailed design included sub-system specifications calculation, modelling and manufacturing. Flight tests and iterations were conducted to verify and improve the aircraft's performance.

The GRIFFIN is a single motor tractor with a tricycle landing gear formation, with a high rectangular wing and conventional thin-plate backward-tapered tail on a truss-structured fuselage. The single motor was selected to reduce the overall weight of the aircraft. The tricycle landing gear was chosen for providing larger ground clearance considering M3 of banner towing, and better maneuverability. The high wing was opted to optimize payload and avionics compartment design. The conventional tail was selected due to structural simplicity and excellent stability characteristics. Further, the team targets to carry 3 passengers, 1 cargo in M2 to maximize aircraft speed, and a 13-foot long, 2.5-foot-wide banner in M3 to maximize banner-sizing score. The Griffin was constructed out of plywood, balsa, and CFRP strips and rods, while the banner mechanism was made of light-weight tricot fabric.

Properties	M1	M2	M3	GM
Payload (lbs)	N/A	0.509	N/A	N/A
Banner (lbs)	N/A	N/A	0.551	N/A
MTOW (lbs)	5.29	5.8	5.95	N/A
Cruise Speed (ft/s)	105.0	105.0	55.9	N/A
Number of Laps	3	9	5	N/A
Mission Time (sec)	93.75	281.25	293	25

Table 1 Predicted Performance of the GRIFFIN

2 Management Summary

The HKU DBF team is an entirely student-managed team from HKU consisting of 17 members, 6 of whom are seniors and the rest underclassmen. The team is financially supported by the HKU Innovation Academy, HKU Mechanical Engineering Department, Engineering Faculty, and by various grants and sponsorships. Likewise, the team is technically advised by alumni and academics in periodical design reviews.

2.1 Team Organization and Roles

The HKU DBF technical team are divided into 3 categories, where green indicates academics and pilot; blue indicates the team leader, 6 technical divisions and 1 helper team, and yellow indicates administrative division as seen in Figure 2.1. The team leader manages the team's societal functions like member intake, events, and direct communication with the Engineering Faculty and sponsorships. The leader also works on schedule planning with the project manager, technical administration with the administrative officer, design with the design officer and finance with the financial officer.

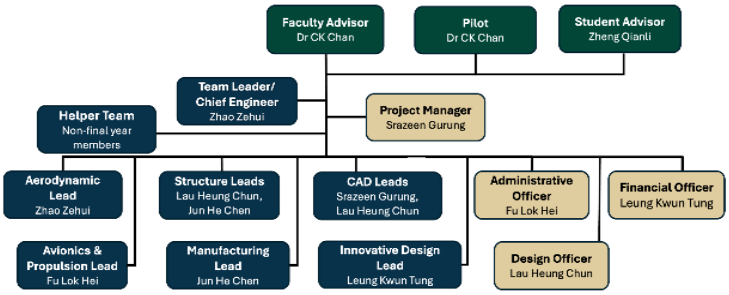


Figure 2.1 Team Organization and Roles

On the technical side, each sub-team has a team lead overseen by the chief engineer who manages the overall project. The chief engineer is responsible for maintaining and tracking the project timeline and communicates directly with the team leads setting up key milestones, meetings, and deliverables. All technical leads manage logistics and organize design reviews with the supervisor. Project deliverables are set out by the chief engineer with all technical leads during regular design reviews and delegated amongst helpers in the team based on individual skills and interests. Members work closely with their team leads who report back to the chief engineer on deliverable updates. Any complications that arise or major design decisions that cause progress delays result in a full team meeting to analyze the advantages and drawbacks of the subsequent steps forward in the design. Table 2.1 shows the roles and responsibilities of each division.

Team	Tasks	Skills
Aerodynamics	Aerodynamics analysis for sizing and desired performance	Fundamental Aerodynamics, derive parameters, CFD simulation
Propulsion & Avionics	Motor and propeller selection, electronics selection for stability and control	Analysis of required and available power; electronics, stability and control theory
Structure	Design airframe and detail structure and conduct testing to ensure structural stability	Structural analysis and FEA knowledge
CAD	Model design and modification	CAD and FEA knowledge
Manufacturing	Manufacturing the aircraft	Machine tools and material property analysis.
Innovative Development	Testing on various innovative materials	Knowledge on 3D printing techniques and 3D print materials

Table 2.1 Team Roles and Individual Skillsets

2.2 Schedule and Major Milestones

The Gantt Chart in Figure 2.2 is planned by the Chief Engineer to oversee the progression of the team and the timeline for objectives and milestones related to the development of the GRIFFIN and designed by Project Manager to provide clear visualization for the team. The Gantt Chart structures the development around three primary prototypes. Prototype 1 (V1) was designed to validate the maximum passenger/cargo configuration (48:13 ratio). Following flight testing, insights from V1 informed the strategy for Prototype 2 (V2), which pivoted to focus on the minimum viable configuration to maximize speed and laboratory space. Prototype 3 (V3) serves as a refinement of V2, with specific improvements to the cargo compartment and banner release mechanism. Whilst the prototyping strategy has been sound, it should be noted that the manufacturing and testing phase of Prototype 1 took considerably longer than anticipated. This delay has directly impacted on the original schedule, leading to the cancellation of the planned Prototype 4 and Prototype 5 to ensure the team can focus resources on the remaining objectives and the final design report.

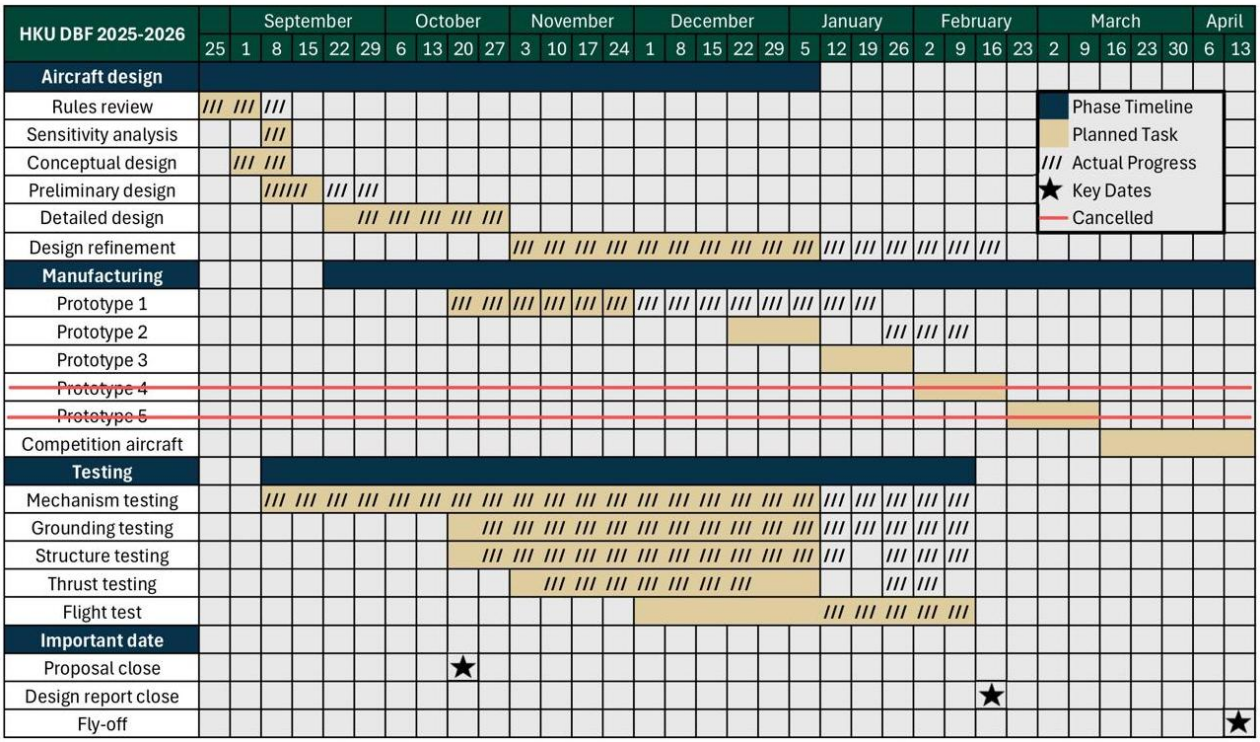


Figure 2.2 Gantt Chart with Planned Task and Actual Progress

3 Conceptual Design

The team analyzed the competition rules to identify the key performance parameters that would maximize the overall flight score during the conceptual design phase. The 2025–2026 AIAA DBF competition rules [1] require teams to develop a solution capable of completing four missions centered on a banner-towing bush plane. The Ground Mission (GM) focuses on the rapid assembly of the aircraft into its flight-ready configuration, including the installation of passengers and cargo for Mission 2 and the banner mechanism

for Mission 3. For the flight missions, Mission 1 (M1) serves as a proof-of-flight demonstration. Mission 2 (M2) requires flight with the payload installed and properly secured. Mission 3 (M3) involves stowage, in-flight deployment, towing, and release of the banner.

3.1 Problem Statement

3.1.1 Mission Scoring

The competition has two main scoring avenues, the written reports and the flight missions. Each mission score is summed together to produce a total mission score, which is then multiplied by the report score to determine the core competition performance. The report score itself is a weighted function of the proposal score and the design report score. Additionally, a participation score is added for attending the competition, passing technical inspection, and attempting the first flight mission. The scoring mechanisms are defined as follows:

$$Total\ Mission\ Score = GM + M1 + M2 + M3 \tag{1}$$

$$Report\ Score = 0.85 \times Design\ Report\ Score + 0.15 \times Proposal\ Score \tag{2}$$

$$Overall\ Competition\ Score = Total\ Mission\ Score \times Report\ Score + Participation\ Score \tag{3}$$

Before any mission can be attempted, the aircraft must pass a comprehensive technical inspection to prove airworthiness and verify compliance with all rules and safety requirements. Once passed, the aircraft enters the staging area for each flight attempt. In the staging box, all propulsion battery packs and payloads are initially removed from the aircraft. The ground crew member is then given a strict 5-minute time limit to assemble the aircraft into its flight configuration, installing batteries and payloads specific to the intended mission. Each flight mission follows a standardized flight course designed to evaluate aircraft performance and handling. The course consists of two 1000ft straight segments, each followed by 180-degree banked turns to reverse direction. After completing the second straight segment and turn, the aircraft performs a 360-degree turn before returning to the start-finish line. To successfully complete a lap, the aircraft must cross the start-finish line in the air. While the landing itself is not counted within the mission time window, a successful landing on the runway is required for the mission to receive a score.

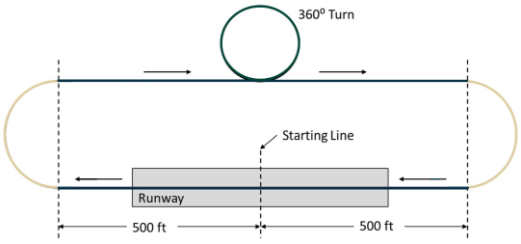


Figure 3.1.1 Flight Course

3.1.2 Ground Mission

The Ground Mission is a timed demonstration that evaluates the efficiency of the aircraft's design for rapid reconfiguration between mission configurations, with the propulsion battery removed. The scoring equation is as follows, where the team's time is normalized against the fastest time achieved by any competing team.

$$GM = \frac{Time_{min}}{Time_{Griffin}} \tag{4}$$

The mission consists of three phases. Phase one requires loading all passengers and cargo with required restraints, followed by the crew member returning across the start-finish line and the pilot demonstrating active flight controls. Phase two requires removing all passengers and cargo, then installing the banner in its compactly stowed configuration, followed by another flight controls demonstration. Phase three requires the crew member holding the aircraft vertically with tail down while the pilot remotely demonstrates banner deployment followed by release. Throughout all phases, the aircraft must remain on its landing gear in the upright configuration and may not be picked up or rotated during payload handling. Only the assembly crew member may touch the aircraft, the propulsion battery must remain removed, and flight controls must be verified active after each loading and unloading phase. After successful completion, teams receive one additional attempt to improve their score.

3.1.3 Mission 1 – Test Flight

Mission 1 requires three laps within five minutes with no payload. Time starts at first throttle advance for take-off. A successful landing is required. Successful completion earns 1.0 point.

$$M1 = 1.0 \tag{5}$$

3.1.4 Mission 2 – Charter Flight

Mission 2 is a charter flight where teams select the number of passengers and cargo to carry, up to the maximum declared during technical inspection. The score is based on net income generated during the flight within a 5-minute window. The team's net income is normalized against the highest net income achieved by any competing team, with the resulting value added to a base score of 1.0.

$$Net\ Income = Income - Cost \tag{6}$$

$$Income = \Big(\#Passengers \times \big(LP1 + (LP2 \times \#Laps)\big)\Big) + (\#Cargo \times \big(LC1 + (LC2 \times \#Laps)\big)) \tag{7}$$

$$Cost = (\#Laps \times \big(Ce + (\#Passengers \times Cp) + (\#Cargo \times Cc)\big) \times EF \tag{8}$$

$$EF = Efficiency\ Factor = \frac{Total\ Propulsion\ Battery\ Capacity\ (W-hrs)}{100} \tag{9}$$

$$M2 = 1 + (\frac{Net\ Income_{Griffin}}{Net\ Income_{Max}}) \tag{10}$$

3.1.5 Mission 3 – Banner Flight

Mission 3 requires in-flight deployment, towing, and release of a banner within a 5-minute window. The banner must be stowed externally, remain vertical during towing, and display the team’s university name or logo on both sides.

$$RAC = 0.05 \times Wingspan + 0.75 \tag{11}$$

$$M3 = 2 + \frac{(\frac{(\#Laps \times Banner\ Length)}{RAC})_{Griffin}}{(\frac{(\#Laps \times Banner\ Length)}{RAC})_{Max}} \tag{12}$$

3.2 System Design Requirements

As outlined in [1], the aircraft's general requirements, M2 and M3 requirements, are outlined in Table 3.2, which guided the team in the preliminary design analysis and final design of the GRIFFIN. Compulsory requirements and preferred desirable characteristics are labelled as "-C" and "-D" respectively.

Category	Label	Design Requirements	Req. No
Main Aircraft	Structures	The airplane maximum wingspan is 5 feet.	AC-1.1
		All structure, components, and payloads must be mechanically fastened.	AC-1.2
		Aircraft must be strong enough to handle payload weight.	AC-1.4
	Aerodynamics	Aircraft must achieve stable level flight in all flight missions.	AC-2.1
		Aircraft should maximise L/D to maximise efficiency.	AD-2.2
		Aircraft wingspan must be a maximum of 5 ft.	AC-2.3
		Aircraft must be able to fly at an altitude of 165ft AGL.	AC-2.4
	Propulsion	Total stored energy of the propulsion battery ≤ 100Wh	AC-3.1
		The propulsion system must support the plane flying for 5 minutes	AC-3.2
		The maximum current rating of an arming fuse is 100 amps.	AC-3.3
	Performance	The aircraft must have sufficient T/W and W/S to take-off within the runway length of approximately 152.4 m (500 ft).	AC-4.1
General	Payload	The airplane must be capable of carrying a minimum number of three passengers and one cargo.	GC-1.1
		The passengers must be carried in one single passenger compartment on a single plane parallel to the horizontal plane while in flight.	GC-1.2
		There must be a separate compartment for passengers and cargo by a solid bulkhead and can only be accessed by one door for each.	GC-1.3
		Each door or hatch must remain rigidly attached to the airplane when opening and in the open configuration.	GC-1.4
		The passengers must be fully restrained in a vertical and upright orientation and prevent movement during all phases of flight.	GC-1.5
		The cargo must be carried in a separate cargo bay(s) that may be in front of, behind, or below the passengers.	GC-1.6
Banner	Structures	The banner mechanism must perform stow, tow, deploy and release event.	BC-1.1
	Performance	The aspect ratio of the banner must be within 5 and the minimum length of the banner must be 10 inches.	BC-2.1
		The banner must be remotely deployed in flight and remain in the vertical orientation position.	BC-3.1
		The banner should be deployed at the first downwind direction and released at the last lap.	BD-4.1

Table 3.2 Subsystem Design Requirements

3.3 Sensitivity Analysis

This sensitivity analysis aims to evaluate the influence of key design and mission parameters on the performance of an aircraft in various mission scenarios. Sensitivity analyses were conducted on MATLAB [2] and Excel [3] by varying parameters such as the number of passengers, cargo number, and battery

capacity for Mission 2 (M2) as well as the lap number, banner length, and RAC for Mission 3 (M3). The sensitivity of each parameter was then assessed using a graphical representation of parameter variations, their corresponding impact on performance metrics, and the steepness of the expected trends. The Ground Mission (GM) score is determined by comparing the team's mission completion time against the fastest time achieved across all competing teams. To quantify this in a sensitivity analysis, the team established a baseline benchmark with the fastest possible GM time assumed to be 25 seconds corresponding to the minimum payload configuration of 3 passengers and 1 cargo. For Mission 1 (M1), 1 point will be awarded after a successful mission. For the purposes of the sensitivity analysis, Mission 1 is assumed to always be completed.

For M2, all parameters increase linearly with the Mission 2 score, except for battery capacity. The number of laps has the steepest gradient among all parameters, meaning this is the most sensitive factor for Mission 2's scoring. This is followed by the number of cargoes, number of passengers and battery capacity. Interestingly, the team needs to demonstrate the plane's capability to carry at least 3 passengers for each piece of cargo in the technical inspection. To determine which parameter to prioritize, the team had to account for the effect of changing the number of passengers and cargos simultaneously rather than changing the two variables individually. By combining the two graphs, an increase of 50% in the number of cargoes and passengers yields a 6.99% higher score compared to a 50% increase in lap count. Therefore, the team chose to maximize the number of payloads in prototype 1. In prototype 2, the team chose to maximize the number of laps by building a smaller aircraft to see its effect on the mission score.

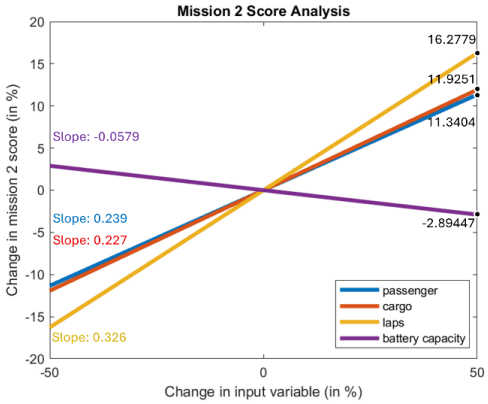


Figure 3.3a Mission 2 Score Analysis

For M3, both the banner length and number of laps have the same linear relationship with the M3 score when independently varied, so maximising either would be beneficial, although admittedly, increasing the banner length is more readily feasible if the team is willing to accept the penalty in increased drag. Since the team has chosen the lightest configuration of 1 cargo and 3 passengers in prototype 2, the focus has been to maximise the aircraft speed, and thus the number of laps, while maintaining a considerable banner length. This banner length was chosen as a compromise that is not so short that it lowers the overall score but not too long that it induces too much drag and negates the effect of completing more laps. The other factor affecting M3, the Rated Airplane Cost (RAC), is governed by the aircraft's wingspan and has a relationship with the M3 score that is best described with a quartic regression. This relationship

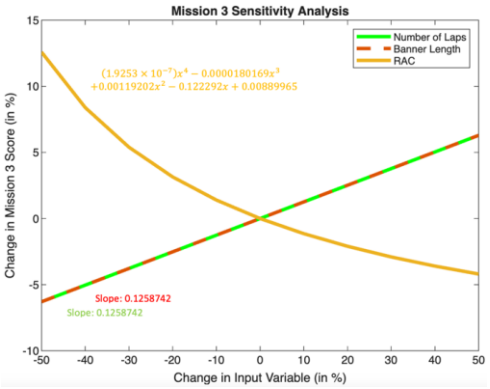


Figure 3.3b Mission 3 Score Analysis

states that with a lower wingspan and RAC, the score shoots up considerably. With the above observations in mind, the team lowered the wingspan of the aircraft from 60 inches in the first prototype to 49 inches in the final design to lower the RAC from 1 to 0.954 while adhering to the competition requirements of achieving a minimum RAC of 0.9. With this RAC and wingspan, the team is confident in towing a banner of 156 inches at a cruise speed of 55.9 ft/s to complete at least 5 laps in M3.

3.4 Aircraft Configurations Decision Matrix

The characteristics of the aircraft configurations are identified and prioritized based on the 5 steps of selection process as shown in Table 3.4.

No.	Step	Description
1	Identify Characteristics	The most desirable characteristics of the configuration choice are identified.
2	Weighting	The characteristics are given a weighting out of 100 (most important) based on their impact to mission performance.
3	Identify Options	Potential suitable configuration choices are examined further based on research and team experience.
4	Scoring	The potential choices are scored out of 1 (best performance) based on how well they perform in relation to each characteristic.
5	Total Score	The total score for each choice is calculated as the sum of the scores for the characteristics multiplied by their respective weights, with the choice of the highest total score being selected. The total score formular is shown below: Total Score = (Weight 1) x (Factor 1) + (Weight 2) x (Factor 2) + (Weight 3) x (Factor 3) ...

Table 3.4 Configuration Selection Process

3.4.1 Main Aircraft Wing Configuration

The configuration considered for the wing planform was swept-back wing, rectangular wing and tapered wing as seen in Table 3.4.1a. The score given to each wing planform ranges from -1 to 1, where -1 is the least effective choice. While both swept back and tapered wings provide better stability, swept back wing reduced the most weight compared to rectangular and tapered wings, but was also the most time-costly to manufacture of all the wing planforms. With reference to Gudmundsson [4], a rectangular wing provides better aerodynamic efficiency than that of the other two planforms under low Re flight (less than 1×10^6).

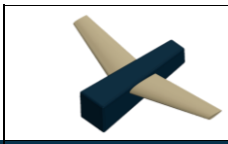
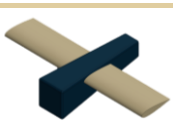
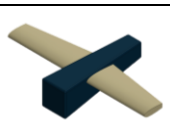
				
Wing Planform	Weighting	Swept Back	Rectangular	Tapered
Manufacturability	20	0	1	0
Aerodynamics	20	0	0.5	0
Flight stability	25	1	0.5	1
Lift	15	0.5	0.25	0.25
Drag	10	0.25	0.5	0.25
Complexity	10	0	1	0
Total Score	100	35	61.25	31.25

Table 3.4.1a Wing Planform Configuration

The configurations considered for the wing placement were high wing, mid wing, and low wing, as seen in Table 3.4.1b. The high wing configuration was selected due to its superior ground clearance and lateral stability compared to the other configurations, but more importantly, the ease of payload access played a significant factor in the final decision.

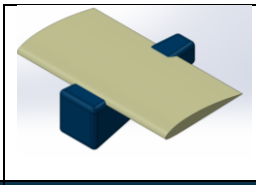
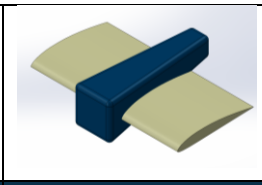
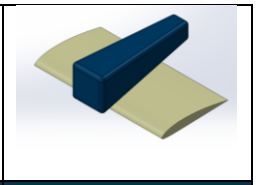
				
Characteristics	Weighting	High wing	Mid wing	Low wing
Ground Clearance	20%	1	0.5	-1
Lateral Stability	20%	1	0	-1
Structural Weight	20%	-1	0	0
Manufacturing Complexity	15%	1	-1	-1
Payload access	25%	1	-1	0
Total Score	100	60	-30	-55

Table 3.4.1b Wing Placement Configuration

3.4.2 Tail Attachment Configuration

The tail attachment configuration was evaluated across three options: conventional, V-tail, and T-tail, using a weighted decision matrix with criteria of manufacturability (15%), weight (25%), structural rigidity (20%), drag (20%), and flight stability (20%).

Conventional tail achieved the highest total score (+0.94 / +94%), excelling in weight, rigidity, drag, and stability due to simple attachment and decoupled control surfaces. V-tail scored +0.58 (+58%), benefiting from lower weight but penalized by complex fuselage integration and coupled pitch-yaw response. T-tail ranked lowest (+0.16 / +16%), suffering from increased weight and structural stress at the horizontal-vertical junction, with minimal aerodynamic advantage at dedicated flying speed.

Final selection: Conventional tail layout. This configuration provides the best overall balance of performance, manufacturability, and practicality for Mission 3 banner towing stability and repeated flight testing.

				
Empennage	**Weighting**	**Conventional**	**V-tail**	**T-tail**
Manufacturability	15	0.9	0.2	0.4
Weight	24	1.0	0.8	-0.3
Structural Rigidity	20	1.0	0.7	0.6
Drag	20	0.9	0.6	0.4
Flight Stability	20	0.9	0.5	0.6
Total Score	100	94.0	58.0	16.0

Table 3.4.2 Tail Attachment Configuration

3.4.3 Main Aircraft Tail Configuration

The empennage configuration was evaluated across four distinct planform shapes: rectangular, forward-tapered, aft-tapered, and trapezoidal. A decision matrix was constructed, taking into account the following weighted criteria: manufacturability (12%), structural weight (24%), flight stability (36%), and drag characteristics (28%).

Rectangular scored lowest (+26%) due to poor drag and stability despite excellent manufacturability and weight. Forward-tapered achieved the highest score (+78%), offering the best balance of low induced drag, superior lift distribution, and good stability for low-speed banner towing. Aft-tapered and trapezoidal ranked in between (+72% and +69%).

Final selection: Forward-tapered planform for both horizontal and vertical stabilizers, maximizing Mission 3 performance while maintaining acceptable weight and manufacturability.

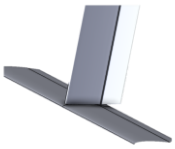
				
Empennage	**Weighting**	**Rectangular**	**Tapered**	**Trapezoidal**
Manufacturability	12	1.0	0.9	0.60
Weight	24	1	0.8	0.75
Flight Stability	36	-0.4	1	0.8
Drag	28	-0.7	0.75	0.65
Total Score	100	28.4	81.2	67.8

Table 3.4.3 Main Aircraft Tail Configuration

3.4.4 Main Propulsion Configuration

For single-motor aircraft, there are 3 common configurations: tractor (motor at the front), pusher (motor at the back), and top-mounted (motor at the top). For dual motor aircraft, the most common configuration is wing mounted. The above four configurations are being compared according to manufacturability, weight, stability, drag, ground clearance, propellor efficiency, and noise and vibration. A tractor design is selected for its ease of manufacturing, favorable balance between stability and aerodynamic drag, and sufficient ground clearance.

					
Motor	**Weighting**	**Tractor**	**Pusher**	**Wing Mounted**	**Top Mounted**
Manufacturability	10	0.7	0.2	0	0
Weight	15	0	0	-1	-0.2
Stability	20	0.6	-0.5	0.3	-0.5
Drag	20	0	1	0.5	1

Ground Clearance	10	0.8	−1	0	1
Propeller Efficiency	15	0.8	0.6	0.5	1
Noise and vibration	10	0.6	0.4	0.3	0.2
Total Score	100	45	15	11.5	34

Table 3.4.4 Main Propulsion Configuration

3.4.5 Main Landing Gear Configuration

A trade study was conducted to select the optimal landing gear configuration for the AIAA competition. The tricycle landing gear achieved the highest score of 0.525 due to its superior performance across mission-specific critical criteria.

While the taildragger configuration offers slight advantages in weight and cost, taildraggers are prone to ground looping and exhibit dangerous directional instability during take-off; as the tail lifts, the centre of gravity shifts ahead of the main wheels, requiring constant rudder correction to counteract the swerving induced by propellor forces.

The tricycle gear was, thus, selected as the most balanced and reliable solution, providing exceptional stability during take-off and landing, intuitive and predictable ground steering essential for precise operations, and a structurally straightforward attachment point for the banner towing mechanism at the rear fuselage to minimize interference during the ground mission.

					
Landing Gear Type	**Weighting**	**Taildragger/Conventional**	**Tricycle**	**Bicycle/Tandem**	**Quadricycle**
Ground Handling/Steering	15	-0.25	0.75	-0.5	0.25
Landing Stability	15	-0.5	0.75	-0.25	0.5
Banner Towing Integration	15	-0.25	0.5	-0.5	1
Weight/Drag	20	0.5	0.25	0.75	-0.25
Cost	20	0.75	0.5	0.25	-0.5
Design and Manufacturing Simplicity	15	0.75	0.5	0.25	-0.25
Total Score	100	21.25	52.5	5	7.5

Table 3.4.5 Landing Gear Configuration

3.4.6 Fuselage Configuration

Comparing the first prototype, which aimed to carry the highest possible payload, to the second prototype, which prioritized increased lap time with minimal payload, the fuselage architecture underwent significant evolvement from a double-deck to a single-deck design driven by ease of manufacturability, weight reduction through elimination of the middle panel and associated structural members and decreased cross-sectional area. In practice, the avionics components interfered considerably with the upper passenger

compartment deck in the first prototype. Consequently, the top wing-seat latch and rear latch for passenger access were replaced with side latches, while the cargo compartment latch remained as side-access. The retaining mechanism itself retained similar functionality to the first prototype, which will be discussed in following chapters. Overall, the fuselage architecture transitioned to a single-deck configuration with all latches positioned on the side doors rather than on top, eliminating interference with the avionics mounted above. Complete solid layers now separate the avionics bay from the passenger and cargo compartments below. This revised configuration forms the basis for the subsequent decision matrix evaluating payload strategy.

3.4.6a Passenger Interior Configuration

The passenger interior configuration was selected based on three criteria: cross-sectional area fuselage area, fuselage length, and loading efficiency, with cross-sectional area being most critical due to its direct influence on parasite drag. Among the three configurations considered—three-abreast, two-one abreast, and one-abreast—the three-abreast arrangement was heavily penalized as it would require approximately seven inches of width including retention, which is inefficient for a low-payload, high-speed design. While the one-abreast configuration requires the longest longitudinal length to accommodate three ducks in series, this increase remained within acceptable limits and even provided beneficial fuselage length for empennage effectiveness and stability. Regarding loading efficiency, the one-abreast configuration excels with the side-latch access design, as each duck can be loaded individually against the compartment wall without requiring alignment between multiple ducks per row. Furthermore, the one-abreast arrangement integrates seamlessly with the pushing latch restraint mechanism by presenting a simple, uniform interface that eliminates sloping surface complications. Based on these advantages, the one-abreast configuration was selected. In the table below, the three configurations are illustrated with the aircraft nose facing left.

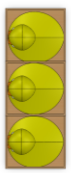
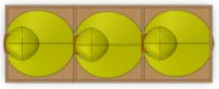
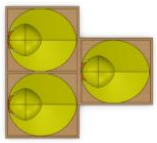
				
Passenger Interior Configuration	**Weighting**	**Three-Abreast**	**One-Abreast**	**Two&One-Abreast**
Fuselage Cross-sectional Area	50	-1	1	0
Fuselage Length	10	1	0	0.5
Loading Efficiency	40	0.5	1	0
Total Score	100	-20	90	5

Table 3.4.6a Passenger Interior Configuration

3.4.6b Cargo Interior Configuration

In terms of the cargo interior configuration, three orientations were considered: Configuration A with the rounded edge facing upward parallel to the longitudinal axis, Configuration B with the hockey puck flat

surface facing upward, and Configuration C with the rounded edge facing upward perpendicular to the longitudinal axis.

These configurations were evaluated against three criteria using a decision matrix, with all cargo accessed via side doors as previously established. Although Configuration B requires the greatest fuselage width, it offers superior structural rigidity by itself due to the large flat contact surface generating a portion of friction for part of the retention. Configuration C demands the same width but requires more complex structural provisions due to reduced contact area and corresponding friction. Configuration B also excels in loading efficiency, which is of the greatest importance. In this sense, practical loading operations prove more difficult for Configuration A. Given that only one hockey puck will be carried and the avionics compartment shares a similar width requirement, the width penalty of Configuration B does not significantly impact the overall design a lot. Based on these advantages, Configuration B with the flat surface facing upward was selected as the optimal cargo interior configuration. In the table below, the three configurations are also illustrated with the aircraft nose facing left.

				
Cargo Interior Configuration	**Weighting**	**A**	**B**	**C**
Fuselage Cross-sectional Area	40	1	0	0
Ease of Structural Requirement	10	0.5	1	0
Loading Efficiency (Through A Side Latch)	50	0	1	1
Total Score	100	45	60	50

Table 3.4.6 Cargo Interior Configuration

3.4.7 Banner Mechanism Position

Three positions of the banner mechanism are considered. The front position was the most desirable position as it is the closest to the CG of the aircraft. Though, it will affect the landing gear installation. Back position created bigger moments for the aircraft which cause negative impact to the stability of the aircraft. Thus, the middle position would be the ideal position for deploying mechanisms.

				
Release	**Weighting**	**Front**	**Middle**	**Back**
Manufacturability	25	0.5	0.75	0.25
Complexity	20	0.25	0.75	0
stability	20	0.75	0.5	0
Release Success	35	0.75	0.75	0.75
Total Score	100	56.25	70	32.5

Table 3.4.7 Banner Mechanism Position

4. Preliminary Design

The preliminary design was an iterative approach used to optimize the engineering design for the main aircraft and the banner mechanism. A combination of simulations, sensitivity analyses, and trade studies in the conceptual design stage shaped the final preliminary design. The specific methods used are outlined in the following sections.

4.1 Design Methodology

The team's design methodology is built on previous HKU DBF techniques and new techniques learned from coursework and academic research. Structured design and analysis were adopted to develop the final aircraft design shown in Figure 4.1. From the sensitivity analysis in section 3.3, it is determined that M2 had the highest score variation through maximizing the payload carried, and the score variation in M3 through maximizing the banner size and minimizing the RAC. The initial empty weight and design details were decided upon based on previous similar competition aircraft. These parameters shaped the first prototype of the GRIFFIN. These requirements further narrowed the design choices into power and wing loading. Aerodynamic analysis started with airfoil selection and geometric parameterization of the wing, tail and fuselage which were simulated in XFLR5 [5], and stability was obtained and compared with previous aircrafts. Various motors and batteries were then selected, and propulsion testing was conducted to obtain realistic data, which were put into MATLAB to predict aircraft performance. Trade studies were conducted to optimize for the configuration that produced the best score under the stated requirements and constraints. Once the preliminary design met all stated requirements in 3.2, a SolidWorks model [6] was then produced to verify the mechanisms, which would be manufactured through laser-cutting and 3D printing. The team planned to perform iterative manufacturing and testing based on experience from the first prototype, with various combinations and constraints considered. Two aircraft prototypes were planned to be manufactured and tested.

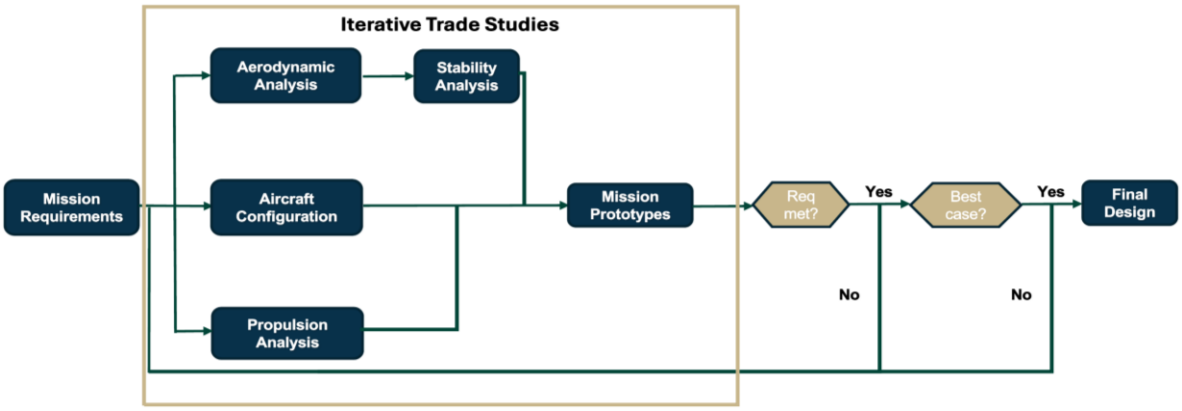


Figure 4.1 Structured Design and Analysis Diagram

4.2 Design Trade Studies

4.2.1 Mission Model

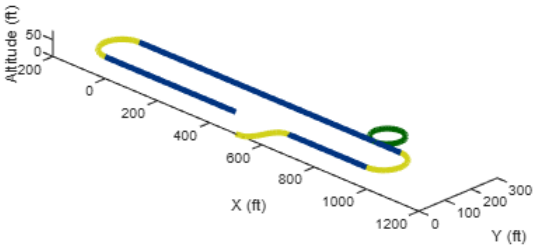


Figure 4.2.1 Mission Model

The GRIFFIN has taken reference to Durmuş [7] and the initial length of the fuselage is taken as 85% of the wingspan which provided the primary estimation of the size and weight of the aircraft. The team has investigated the weight of the aircraft in previous AIAA DBF aircraft design as a reference to the GRIFFIN which is shown in table 4.3.1. The aircraft performance was estimated as a function of empty weight, MTOW, wing area, wing aspect ratio (AR), and other variables relating to lift and drag equations introduced later in section 4.3.1 for each of the three flight missions. The various constraints and assumptions for these variables were drawn from the team's past aircraft and the 2025-2026 competition rule document. The parameters related to lift and drag equations were taken from historical data and XFLR5, which will be verified and updated by testing using a similar aircraft. The mission model shown in Figure 4.2.1, was computed on MATLAB with equations mentioned in section 4.3.1. The sizing studies are determined by the aerodynamic performance during the take-off/climb, cruise, and land stages.

4.2.1.1 Take-off/Climb

At the take-off/climb stage, the throttle is set to 100%, with a ground roll friction coefficient of 0.6 between rubber and concrete (dry) material [8]. The aircraft accelerates to take-off velocity where it pitches up to reach the required α until sufficient lift is produced to accelerate into the air. The aircraft then maintains full throttle and the required α until it reaches the desirable altitude which ends the climb stage. For M1/2, the desirable altitude is 65ft and for M3, it is 100ft to allow more reaction time for the pilot for sudden weather changes during flight. Although take-off distance is not limited this year as mentioned in AIAA 2026 rules [1], the team considered STOL to reduce friction induced in the ground roll process and required T/W to achieve such performance.

4.2.1.2 Straight Flight

In Straight flight, the throttle is adjusted dynamically to achieve the desirable airspeed by using 80% battery usage at maximum, while 20% is reserved for landing. α is adjusted to achieve level flight. If more than 80% of the battery is used, the configuration is invalidated.

4.2.1.3 Turns/Loop

In turns/loop stage, the throttle is adjusted dynamically to achieve the desirable air speed for the turns. α will be adjusted accordingly, depending on the weather change and the target. It increases until the maximum acceptable angle is achieved which includes sufficient room for bank angle to produce upwards lift to balance the weight of the plane. This approach includes a simulation based on 2D simulation. Though, this neglect impacts by complex situations, such as losses of thrust with an inconsistent shift of α due to

weather changes, impact on lift coefficient due to inconsistent velocity, aircraft vibration which affects the aerodynamic performance. The team investigated a Multidisciplinary Design Optimization framework, which includes safety factors and realistic considerations for estimating the aircraft performance, mentioned later in section 4.4.

4.2.2 Airfoil Selection

Reynolds numbers ranging from 200,000 to 1 million were simulated for ten kinds of airfoil in XFLR5, with the results shown on the comparison graph on the software and exported into an excel form which was weighted based on the main requirements set out in the constraints diagram. This included lift coefficient, lift to drag in cruise, climb and turn and the pitching moment. With basic specifications of the main wing, fuselage and empennage specified, three airfoils were taken into 3D space, with data exported and simulated on MATLAB, which calculated the score of M2 and M3. The team investigated a hybrid of 65% CLARK-Y and 35% Wortman FX 63-137 (WFX), and the single use of CLARK-Y, NACA23012 and WFX. The hybrids of WFX and CLK-Y airfoils preserved the good lift creation for flaps and high sensitivity to stall motion in aileron. In the first prototype, the team adopted the hybrid airfoil configuration, although the team has, since then, changed to adopting only the CLK-Y airfoil as it eased the manufacturing process and reduced wing twist issue compared to that of NACA23012 and WFX airfoils.

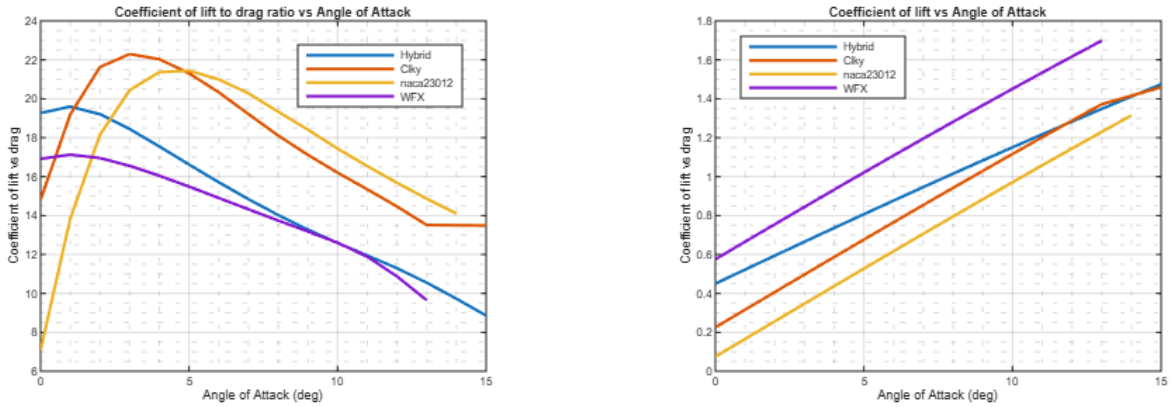


Figure 4.2.2 a) Coefficient of Lift-Drag Ratio vs Angle of Attack and b) Coefficient of Drag vs Angle of Attack

Characteristics	Weighting	Clark-Y	WFX-63 137	NACA23012	Hybrid
CL/CD for turn	30	1	0.5	0.75	0.25
CL/CD for cruise	25	0.5	0.75	0	1
Maximum Lift Coefficient	20	0.5	1	0.25	0.75
Manufacturability	25	1	0.25	0.25	0.5
	Total	67.5	55	33.75	60

Table 4.2.2 Comparison of Airfoil Performance

4.2.3 Propulsion Optimization

The propulsion system was designed to achieve high cruising speed in M2 and high thrust in M3 to overcome the drag induced by the plane and the banner during the flight. First, a thrust-weight ratio of 1.1

was selected to achieve sport-flying performance, meaning the plane should produce at least 3.52lbs of thrust. The following listed the motor-propeller combinations which fulfilled the ratio requirements at a throttle lower than 50%:

Motor	Weight (ozs)	Max Lipo Cell	Power (W)	Remarks	Propeller	Max Thrust (lbf)	RPM	Efficiency (oz/W) at 3.52 lbf
AT3530 Long Shaft KV580	10.51	6s	1400 (Peak)	N.A.	APC 13*8	9.0	9678	~0.22
					APC 14*7	10.60	9496	~0.23
Scorpian A-3017-600kv	5.43 (include accessories)	6s	962 (Peak)	Sport Weight Range: 5.07-11.09 lbs	12x6E	N.A.	10900	N.A.
					13x6E	N.A.	10500	N.A.
SunnySky X3120 V3 KV585	5.61	6s	1625(Max Continuous Power)	Weight for aerobatics: 3.09-3.97 lbs	12x6.5	9.52	11465	~0.21
					13x7	11.11	10500	~0.22

Table 4.2.3 Motor Comparison

Based on the weight, market availability, thrust requirement, and result ground clearance, motor SunnySky X3120 V3 KV585 was chosen. The 13x7 propeller was selected due to its high thrust and relatively high efficiency, which is important when the plane needs to maximize speed and endurance. For battery selection, the team substituted the nominal voltage and the corresponding current to the formula below and allowed a 20% safety margin for preserving battery health:

$$Voltage(V)\ x\ Battery\ Capacity\ (mAh) = Energy\ (Wh) \qquad (4.2.3)$$

Based on the calculation results, a 4500mAh 22.2V battery (E=99.9<100Wh) was chosen. Moreover, dividing the battery capacity by the battery density (~5.67Wh/oz) illustrated that the weight of the battery should be around 1.10lbs. For ESC selection, a safety factor of 1.2 was applied to the maximum continuous current (65A) which means that an ESC with a current rating larger than 78A could be considered. As a result, an 80A ESC was chosen. Moreover, an 80A fuse instead of a 100A blade fuse was chosen to protect the 80A ESC.

4.2.4 Empennage Sizing

The empennage plays a critical role in ensuring flight envelope integrity, maneuverability, and static/dynamic stability. Empennage sizing typically employs two primary approaches: rule-of-thumb methods and empirical formulae [9]. Lennon [9] recommends sizing the tail surfaces as a fixed percentage of the wing area for simplified RC aircraft design. For more precise calculations, the empirical method utilizes the horizontal tail volume coefficient (V_h) and vertical tail volume coefficient (V_v), defined as:

$$V_h = \frac{S_h l_h}{S_w MAC};\ V_v = \frac{S_v l_v}{S_w b}$$

where S_h and S_v are the horizontal and vertical stabilizer areas, l_h and l_v are the respective moment arms, S_w is wing area, MAC is mean aerodynamic chord, and b is wingspan. Both coefficients are directly proportional to aircraft stability and control authority. With wing design parameters (including MAC, l_h, l_v) already determined by the Wing Design Division, the stabilizer areas are solved as:

$$S_h = \frac{V_h S_w MAC}{l_h};\ S_v = \frac{V_v S_w b}{l_v}$$

To ensure adequate stability margins, tail volume coefficients were set to V_h = 0.9 and V_v = 0.07 — values higher than typical ranges (0.6 and 0.04) — providing conservative stability for the banner-towing mission.

4.2.5 Control Surface Sizing

Control surfaces were sized using three stages, Lennon figure 2 [6] for preliminary design, XFLR5 for trim and turn deflection and then a pilot test. The team used historical values from HKUDBF2025 to guide initial sizing with the values shown in Table 4.2.5. This was then tested in an actual flight test to verify reasonable deflection. The team also investigated the use of flaps in prototype 1 to increase the lift force, however removed them in the final design due to a large increase in drag.

	Chord %	Span %
Aileron	25	4.16
Elevator	40	100
Rudder	50	86

Table 4.2.5 Control Surface Sizing

These proportions (40% elevator, 50% rudder) are conservative and larger than typical single propeller aircraft (elevator 25–35%, rudder 30–45%) [9]. The oversized surfaces offer significant advantages in Mission 3: enhanced low-speed response, improved disturbance rejection from banner dynamics, and reduced pilot workload in gusty conditions. While larger control surfaces introduce minor increases in profile drag and hinge moment, the impact on endurance and lap count is limited in this low-speed regime. The stability and control benefits far outweigh the small aerodynamic penalty, making this sizing reasonable and effective for achieving stable banner towing performance.

4.2.6 Payload Compartment Design

The design evolution from the first prototype to the second was heavily influenced by lessons learned during testing and analysis of the first GRIFFIN aircraft, which originally carried 48 ducks and 16 hockey pucks using a closing latch retention mechanism. In the first prototype, two latches were employed for the passenger compartment, one at the wing seat and one at the rear, along with a side latch providing access to the cargo compartments. For the passenger compartment, ducks were placed inside and a latch with foam backing pushed them against the bottom panel as the latch closed, as shown in Figure 4.2.6a. However, loading efficiency proved significantly lower than anticipated, primarily due to the need for frequent adjustments when placing large numbers of ducks. And the size variations among individual ducks, which required additional foam padding added weight and manufacturing complexity. Although payload quantity was the dominant factor in Mission 2 scoring, the ground mission performance suffered due to overestimated loading efficiency. The cargo compartment functioned adequately in terms of restraint but

also suffered from overestimated loading efficiency and required careful handling to avoid damaging the tight-fit structure as shown in Figure 4.2.6b. Additionally, the double-deck configuration designed to accommodate the large payload without excessive fuselage length resulted in significant wasted space on the lower deck, contributing unnecessary weight.

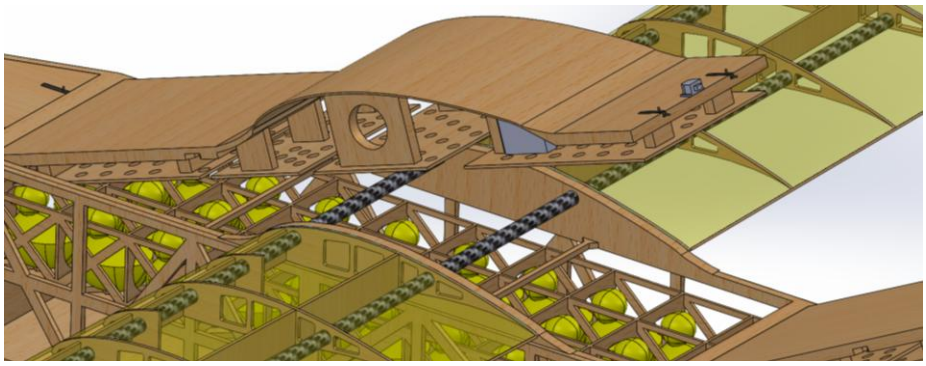


Figure 4.2.6a Prototype 1 Passenger Retainment Mechanism

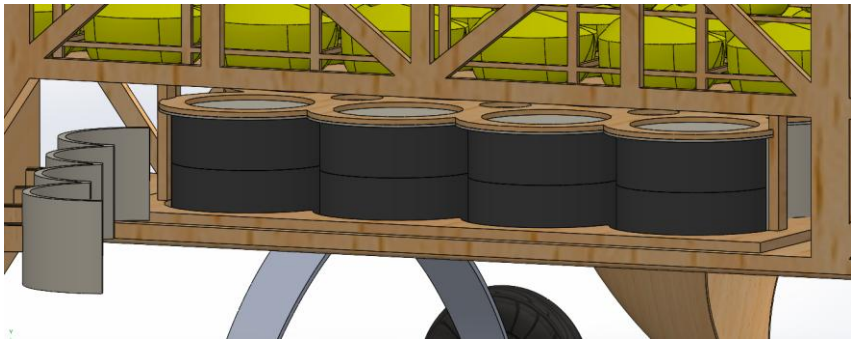


Figure 4.2.6b Prototype 1 Cargo Retainment Mechanism

Based on these findings, the team decided through careful simulation and analysis to abandon the high-payload approach and adopt a simplified configuration of three ducks and one hockey puck. This huge payload quantity transition enabled a single-deck layout with all payloads on one level, dramatically improving space efficiency and ground mission speed. For the duck arrangement, the wing seat latch mechanism was eliminated due to the abandonment of the double-deck fuselage design, reducing complexity. A side latch configuration was adopted instead, offering superior accessibility without interfering with avionics components mounted on top below the wing, such as the receiver, UBEC, and avionics battery. This approach also avoided the need for a complex curved latch matching the wing seat contour as required in the first prototype.

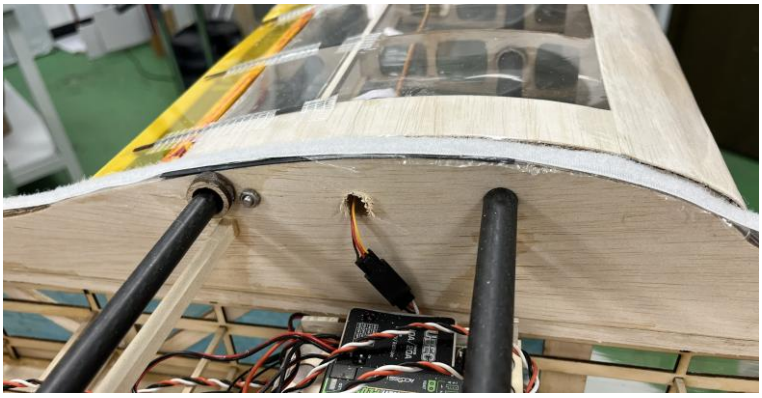


Figure 4.2.6c Prototype 1 Wing Seat Design

Flight testing of the first prototype also revealed significant fluttering issues, attributed to the wing being installed in separate halves rather than as a single unit. This discovery directly influenced the second prototype design, where the wing halves were joined together first and mounted as one piece onto a dedicated wing seat. This high-wing configuration, however, precluded top access to the payload compartment, further reinforcing the side-access design choice. The second prototype therefore featured side doors accessing both passenger and cargo compartments, retaining the closing latch mechanism but now pushing the ducks laterally against the side wall panel rather than vertically against the bottom panel. To separate individual ducks and ensure compact, rigid restraint, wooden strips were used instead of full-height panels as shown in Figure 4.2.6d. Testing confirmed that these strips alone provided adequate separation and rigidity between ducks, offering a weight-saving advantage over the full separating panels used in the first prototype.



Figure 4.2.6d Prototype 2 Passenger Compartment Retaining Mechanism

4.2.7 Banner Sizing and Mechanism Design

Four design mechanisms were considered and tested for the banner stowing, towing and deployment stage. The 4-bar linkage bucket provided strong rigidity to hold the banner; however, it occupied too much space

underneath the fuselage and added more weight to the aircraft. The cage and pillars designs adopted balsa wood and aluminum strips as manufacturing materials; however, the pillars design cannot constrain the axial movement of the banner, and the cage design could not hold the banner tight enough. The tube in the rubber bands design tended to entangle the banner wires during deployment. The gripper mechanism demonstrated the most successful deployment of the banner. Moreover, two release mechanisms were considered. The spool design organized the banner wires but vibrated vigorously during flight. The badge reel design, on the other hand, organized the wires inside the box and restricted the deployment speed to make the overall release process steady.

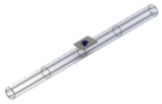
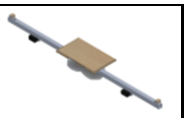
						
Deployment	**Weighting**	Gripper	Cage	Linkage Bucket	Pillars	Rubber bands
Manufacturability	20	0.75	0.25	0.25	0.5	0.5
Complexity	25	0.25	0.5	0	0.5	0.25
Occupation	20	1	1	0.25	0.75	1
Deploy Success	35	1	-1	0.75	-1	0
Total Score	100	76.25	2.5	36.25	-2.5	36.25

Table 4.2.7a Banner Deployment Mechanism

			
Release	**Weighting**	Badge Reel	**Spool**
Manufacturability	20	0.75	0.75
Complexity	25	0.75	0.5
Release Speed	20	1	0.5
Release Success	35	0.75	0.25
Total Score	100	80	46.25

Table 4.2.7b Banner Release Mechanism

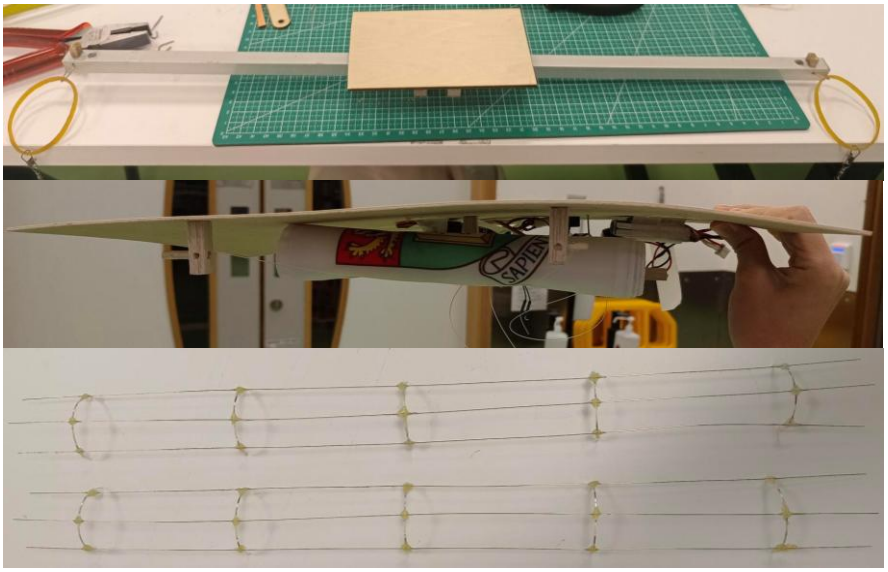


Figure 5.3b) Rubber-band, c) Pin, d) Cage

4.2.8 Banner Material

The team investigated 5 types of banner materials in terms of their impact on drag force, rigidity, and density in Figure 4.2.8. The density of each material is shown in Table 4.2.8, where d) is the lightest material, and a) had the best balance in density and rigidity. The team prioritized a balance in density and rigidity which maintains sufficient steadiness and reduces vortices along the edges.

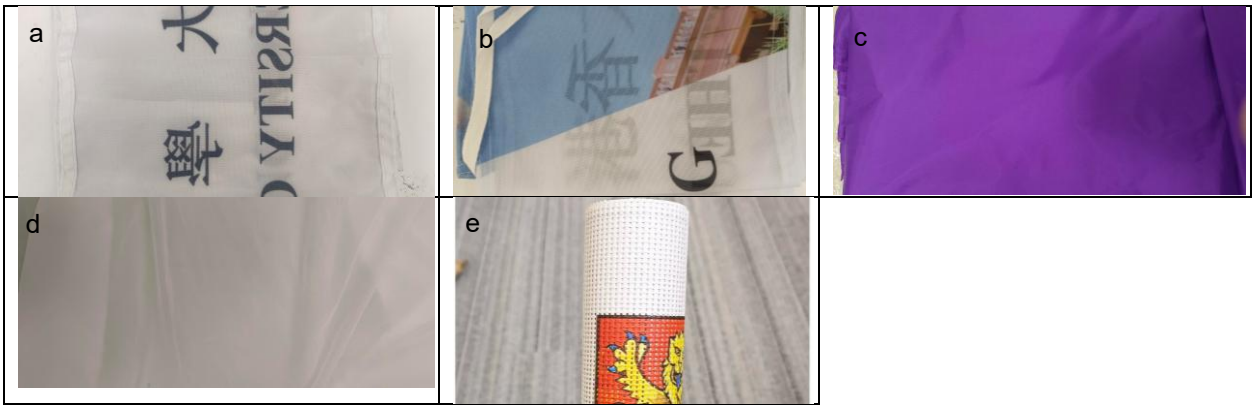


Figure 4.2.8 a) Pongee b) Tricot c) Nylon d) Super light-weight Nylon e) Mesh fabric

Material	Pongee	Tricot	Nylon	light-weight Nylon	Mesh
Density (lbs/ft^2)	0.016	0.019	0.018	0.005	0.079

Table 4.2.8 Banner Material Densities

4.3 Aerodynamic Design Analysis

4.3.1 Lift and Drag Analysis

The lift and the drag coefficient extracted from XFLR5 were initially calculated and updated several times for several prototypes mainly focusing on prototype 1 (where the total weight was estimated to be 16.5 lbs.), and prototype 2 (where the total weight was 3.53 lbs.), values which were input into the Multidisciplinary Design Optimization (MDO) framework. During initial sizing, the targeted C_L was calculated via 4.3.1a from the target mission speeds. The C_D was calculated with the parasitic drag coefficient, aspect ratio, lift coefficient, and Oswald's efficiency factor. The power of a propeller-driven aircraft in steady and level flights can be calculated using Eq. 4.3.1d and 4.3.1e. The aerodynamic center of the aircraft in cruise was estimated as ¼ of the chord length with reference from Glenn Research Center [10].

$$C_L = \frac{2W}{\rho S_W V^2} \quad (4.3.1a) \qquad C_D = C_{D,0} + K{C_L}^2 \quad (4.3.1b) \qquad K = \frac{1}{\pi\, AR\, \epsilon} \quad (4.3.1c)$$

$$D = \left(\frac{1}{2}\rho V^2 S C_D\right) + \left(\frac{kW^2}{\frac{1}{2}\rho S V^2}\right) \quad (4.3.1d) \qquad P = \frac{VD}{\eta} \quad (4.3.1e) \qquad T - D = \frac{MV^2}{R} \quad (4.3.1f)$$

Referenced from the value of previous DBF aircraft in year 2024 and year 2020 [11][12], the initial aerodynamic design of the GRIFFIN was carried out and put into the MDO in 4.3 where the cruise speed and MTOW were particularly focused.

	UNSW2025	UOL2024	USC2020
Cruise Speed (ft/s)	110.0	143.6	147.58
MTOW (lbs)	48.5	14.02	25.44
W/S (lb/ft^2)	5.13	2.13	5.09
Stall Speed (ft/s)	N/A	38.77	56.47

Table 4.3.1 Aerodynamic Design Comparison of previous DBF aircraft

4.3.2 Stability Performance

4.3.2.1 Static Stability

With the data of the empennage sizing as discussed in section 4.2.4, the main wing and the empennage of the aircraft was modelled on XFLR5. Then, the center of gravity (CG) of the aircraft can be estimated using several parameters, including the weight of the individual parts of the aircraft, as mentioned later in section 5.5, weight of the payload, weight of the banner, weight of the avionics/electronic components and the initial estimation of the length of the fuselage and the wing. A static moment was estimated which confirmed that the preliminary CG was ahead of AC during cruise.

Static Margin (%)	CG to LE	AC to LE	Main Landing gear to LE	Nose to LE	Tail to LE
15	-0.133in	-2.062in	-5.61in	+8.477	-22.3in

Table 4.3.2 Distance Measurement for Static Stability

4.3.2.2 Dynamic Stability

The team investigated longitudinal and lateral dynamic stability for cruise stage using XFLR5 [3], A root-locus plot for the complex roots of the modes is shown in Figure 4.3.2.2. A mode is stable if the real part of the root is on the negative side, and the frequency of the oscillation is positively proportional to the magnitude of the imaginary part. The values showed that the aircraft is stable in the roll, spiral and Dutch roll mode. With damping ratio in an acceptable range of short period and phugoid mode, the aircraft are said to be capable of maintaining good stability in both oscillating and non-oscillating flight conditions.

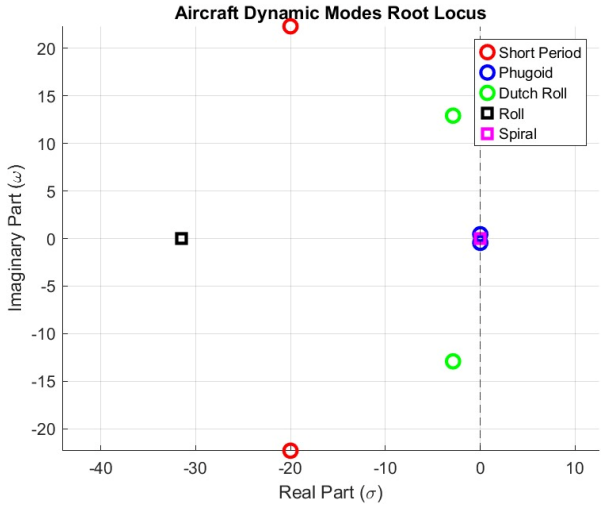


Figure 4.3.2.2a Aircraft Dynamic Modes Root Locus

Mode	ζ	f (Hz)	F_1 (Hz)	t_2 (s)	τ
Short Period	0.668	6.407	4.765	0.193	0.156
Phugoid	0.007	0.0712	0.072	1455	13.90
Dutch Roll	0.217	2.160	2.109	6.900	0.463
Roll	N/A	31.91	N/A	0.022	0.031
Spiral	N/A	N/A	N/A	30.46	N/A

Table 4.3.2.2b Dynamic Stability Data for Longitudinal and Lateral Motion

4.3.3 CFD Analysis

A computational-fluid-dynamic analysis was conducted on SolidWorks [8] to estimate the aerodynamic performance of the aircraft during flight, assuming 105ft/s in cruising. The result of pressure distribution along the aircraft in Figure 4.3.3 shows a room for improvement for aircraft nose and empennage aerodynamic design.

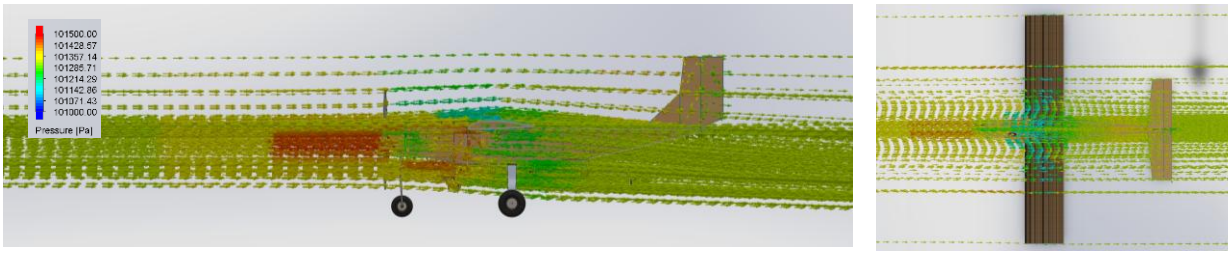


Figure 4.3.3 a CFD Analysis on SolidWorks-Side View, b CFD Analysis on SolidWorks-Top View

4.4 Mission Prediction Analysis

A parametric MDO framework was developed in MATLAB using a deterministic grid-search approach to explore the design space systematically to obtain the predicted performance of the preliminary design that maximizes mission score while satisfying aerodynamic and performance constraints. Several assumptions for the battery selection and empty weight of the aircraft under different conditions were made to provide a realistic scenario in the simulation, with an assumed velocity range based on previous aircraft performances. Then the flight path was estimated by examining the flight motion for three stages: take-off/climb, cruising, landing.

The highest possible score for each individual mission was estimated and was compared with the optimized score case where all missions' requirements were considered to determine the highest score case for the GRIFFIN. For GM, the time for the fastest possible case was estimated as 25 seconds, based on experience from the team. In the iterative optimization simulation, the formula of GM time was written as $GM = 25 + passenger * (n-3) * 8 + cargo * (n-1) * 8$. For M2, the minimum value of the payload is 3PS + 1CR. The performance in M2, including the mass of the aircraft, the

Figure 4.4 Commercial Plane Flight Test



flight time and the number of laps, the flight velocity and the number of payloads, and the battery required were considered. The velocity of the aircraft was estimated to be in a range of 59.06 ft/s to 104.987 ft/s, according to previous aircraft performances. Using an iterative approach, the highest M2 marks were calculated. For M3, the same parameters were estimated These parameters were estimated based on the lift coefficient and the drag coefficient simulated in XFLR5 and wind testing. A value within the range of aspect ratio of the banner from 1 to 5, and a range of banner length from 10in to 157.25in was selected in each iteration. The drag coefficient of the banner and a data reference of velocity and number of laps was taken from a flight test using a commercial plane of 4.41lbs with a 3300 mAh battery and a 6.615 lb-f motor as shown in figure 4.4. Using a similar iterative approach, the highest mark for M3 was calculated. Then, an optimization loop was simulated, by iterating the score calculation and putting them into the score formula provided by AIAA 2025 rules. The simulation went through 100 iterations and thus estimated the top 20 score case for the team. A trade study was then conducted based on the team's constraints and requirements to determine the final prototype design.

4.5 Predicted Mission Performance

From the predicted mission performance simulated on MATLAB, the top 100 score cases are as indicated in Figure 4.5. The bulk group in green color corresponded to score cases where a PS more than 33 and a CR more than 11 were chosen. The light group as orange triangles corresponded to score cases where 1 PS and 3 CR was chosen. The rest of the combinations were indicated as other top cases. The comparison was taken as the ratio of the total score and the aircraft mass, since the size of the aircraft determines the time taken for manufacturing, and the battery and motor selection, and banner mechanism design. Since bulk groups were the highest score cases, the first prototype adopted 48 PS and 16 CR. Through examination after manufacturing and testing process, the light group was selected as the final aircraft design as it took the least time to manufacture and flew faster than other design cases, while the score deviation is mild compared to the bulk group, with a difference of 0.08.

Figure 4.5 Top 100 Case Comparison

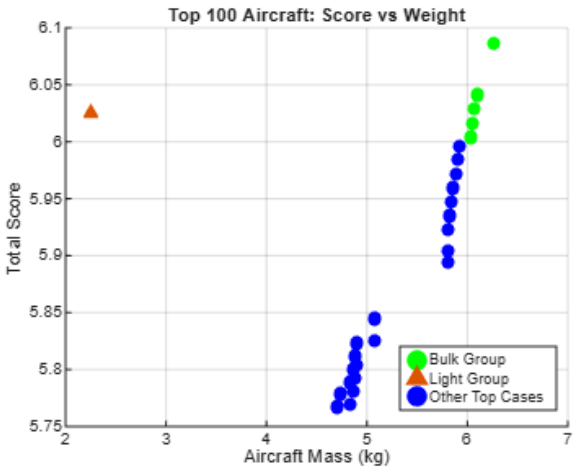


Table 4.5 Predicted Aircraft Performance

Predicted Aircraft Performance	
Total Score	6.026
Wingspan (ft)	4.101
Cargo	1
Passengers	3
M2 V_{cruise} (ft/s)	105
M2 Laps	9
M3 V_{cruise} (ft/s)	55.45
M3 Laps	5
Battery (Wh)	75.50
Banner Length (ft)	12.63
GM Time (sec)	30
M2 Mass (lbs)	4.593
M3 Mass (lbs)	5.906
Banner Aspect Ratio	5.0

4.6 Uncertainties and Assumptions

This score simulation was based on assumptions referenced from previous aircraft designs, and mass properties on CAD assembly while all the data were extracted from previous aircraft experiences and simulations on XFLR5. The battery selection and the weight assumption were based on rough estimation. These may not reflect the actual situation during flight thoroughly due to complex geographical factors in the airfield environment. The highest possible cases for GM, M2 and M3 were estimated based on the team's experience and a range of velocities based on previous aircraft performances which may vary in the future.

5 Detail Design

With the result of the preliminary design, the detail design optimized the structure and payloads of the aircraft to fulfill the desired requirements, with emphasis on reducing weight, manufacturing, and GM time.

5.1 Dimensional Parameters

Table 5.1 shows the final design parameters for the GRIFFIN.

Wing		**Avionics**	
Airfoil	Clark-Y	**Receiver**	Frsky Archer Plus Sr12+
Span (ft)	4.101	**Servos**	Aileron: 2×KM0940MD Elevator: 1×KM0940MD Rudder: 1×KM0940MD Nose Steer.: 1×KM0940MD Banner Deploy: 1×Guohua A0090 Banner Release: 2×Guohua A0090
MAC (ft)	0.668	**ESC**	80A Hobbywing Skywalker UBEC (ESC)
Planform Area (ft°)	2.797	**UBEC**	Hobbywing UBEC 10A-Car
Aspect Ratio	6	**Motor**	
Incidence Angle (°)	0	**Motor**	SunnySky X3120 V3 KV585
Static Margin (%)	10	**Battery**	
Fuselage		**Propulsion Battery**	GAONENG 4500mAh 6S 70C
Length (ft)	1044mm	**Avionics Battery**	ATECH Infinity 550mAh 2S 85C
Width (ft)	92.2mm	**Horizontal Stabiliser**	
Height (ft)	114.5mm	**Tail arm (ft)**	2.13
Vertical Stabiliser		**Span (ft)**	1.8
Tail arm (ft)	2.13	**Aspect Ratio**	4.0
Span (ft)	0.75	**Root Chord (ft)**	0.45
Aspect Ratio	1.50	**Tip Chord (ft)**	0.36
Root Chord (ft)	0.50	**Area (ft^2)**	0.81
Tip Chord (ft)	0.40	**Elevator Chord (ft)**	0.16
Area (ft^2)	0.38	**Elevator Area (ft^2)**	0.29
Elevator Chord (ft)	0.23	**Banner Size**	
Elevator Area (ft^2)	0.17	**Material**	Poly pongee
Propeller		**Length (ft)**	13.12
M1/2/3	EOLO 13x7	**Aspect Ratio**	5

Table 5.1 Final Design Parameters for the GRIFFIN

5.2 Structural Characteristics

5.2.1 Wing Structure

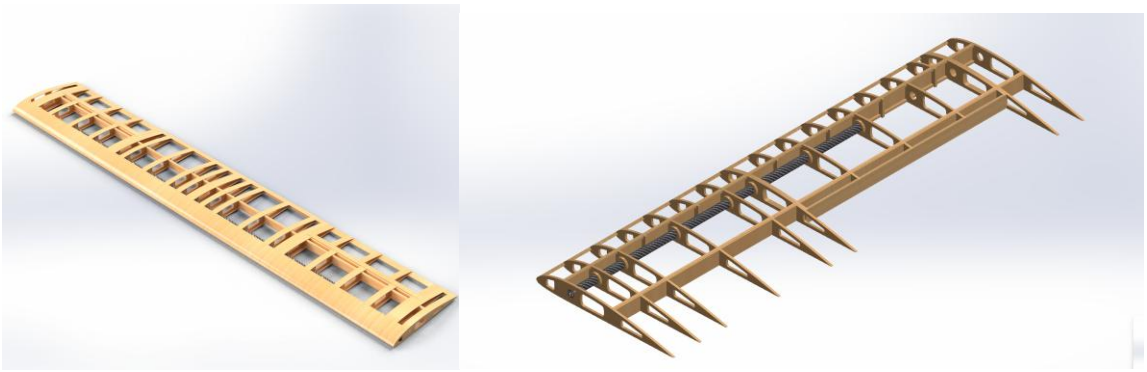


Figure 5.2.1a Wing structure with and without sheeting

As one of the main targets for the competition is to achieve a lightweight structure while maintaining sufficient strength to remain intact throughout the event, material selection was critical. For the wing ribs and aileron, 0.118-in (3 mm) balsa wood was chosen due to its favorable strength-to-weight ratio, with a density of roughly 7.8 lb/ft³ (125 kg/m³), providing a balance between weight savings and structural rigidity. Furthermore, to reinforce the holes for the carbon fiber rods, a 0.394-in (10mm) diameter ring with a thickness of 0.118-in (3mm) was laser-cut and attached around each hole. This reinforcement was validated through finite element analysis; comparing the ribs with and without the ring showed that the local reinforcement reduced the peak von Mises stress at the hole interface from 23.2 psi (1.6e5 Pa) to 4.2 psi (2.9e4 Pa) in **Figure 5.2.1b**, representing an 81.9% reduction in stress concentration. This confirms that the ring effectively distributes the load from the main spar into the rib structure, eliminating the risk of crushing or cracking propagation around the hole. To further save weight, half-ribs were also used in non-critical sections, replacing full ribs where structurally permissible.

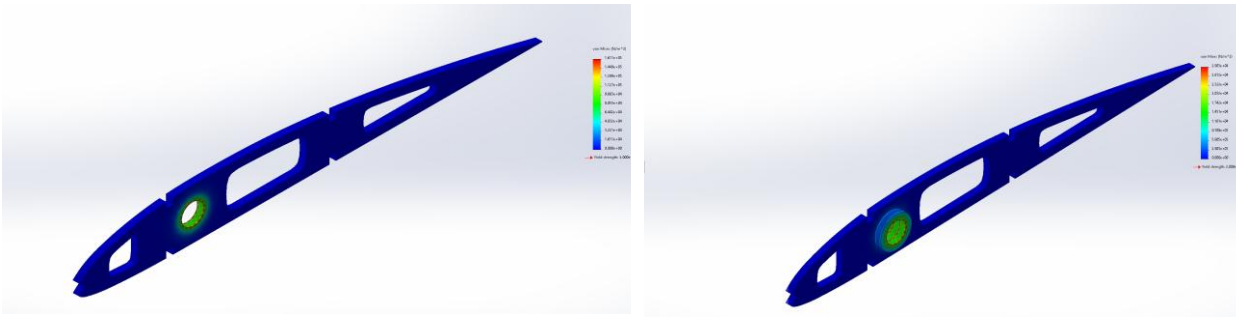


Figure 5.2.1b FEA analysis of the ribs without and with reinforcement rings

The main spar of the wing is a carbon fiber rod with a length of 34.6-in (880mm) for the entire wing and a wall thickness of 0.04-in. A rear spar is located at 62.5% chord to support the aileron hinges. The ribs were spaced evenly along this spar at 1.57-in (40 mm) intervals to maintain the aerodynamic profile.

For the secondary structure, two additional spars were utilized, with webbing on one side to provide further structural support for the main spar. These spars also served a crucial role in aligning the ribs accurately during the wing assembly process. The first secondary spar is located at 20% chord to support the D-box, while the second spar is located at 55% chord for additional support.

To aid with aerodynamics and increase torsional resistance, a 0.039-in (1mm) balsa wood sheet was used to cover the leading edge of the wing, extending from the leading edge back to the first support spar. This forms a classical D-box structure, which significantly increases the wing's torsional stiffness. Furthermore, to increase the stiffness of the remaining wing structure, a vertical strip of 0.039-in (1mm) thick balsa was attached from the end of the D-box to the trailing edge of the wing, effectively forming an I-beam shaped structure along the chordwise direction. The same reinforcement method was applied to the aileron to ensure it maintains its shape under aerodynamic loading.

A limit load factor of 6g was selected based on past team experience and data collected from flight testing. Finite element analysis of the wing structure under ultimate load conditions yielded a maximum von Mises stress of 25,300 psi (174.2 MPa) and a tip deflection of 1.68 in (42.6 mm) over a semi-span of 24.6 in (625 mm), corresponding to 6.8% deflection as shown in **Figure 5.2.1c**. With carbon fiber UTS of 87,000 psi (600 MPa) [13], this gives a minimum Factor of Safety of 3.44. While the deflection ratio is 6.8%, the presence of a stiff D-box and secondary spars ensure adequate torsional rigidity and control authority. The design is deemed structurally adequate for the competition flight.

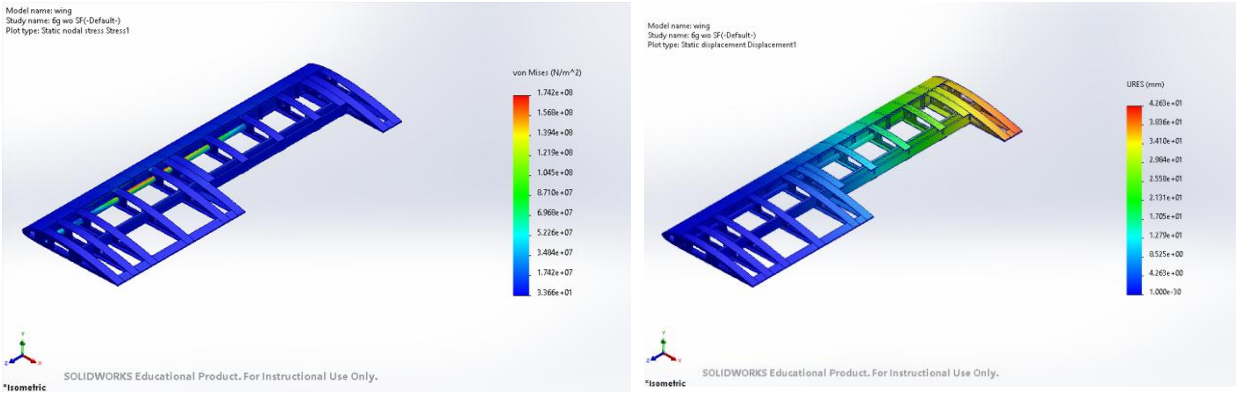


Figure 5.2.1c FEA of Wing Under 6g MTOW Load

To evaluate torsional stiffness, a 7.38 lb-ft (10 Nm) torque was applied to the half-wing model with and without the skin structure (D-box and I-beam) in **Figure 5.2.1d**. The skin reduced tip deflection from 3.49-in (88.6 mm) to 2.12-in (53.8 mm), a 39% improvement in torsional stiffness, validating the effectiveness of the closed D-box configuration.

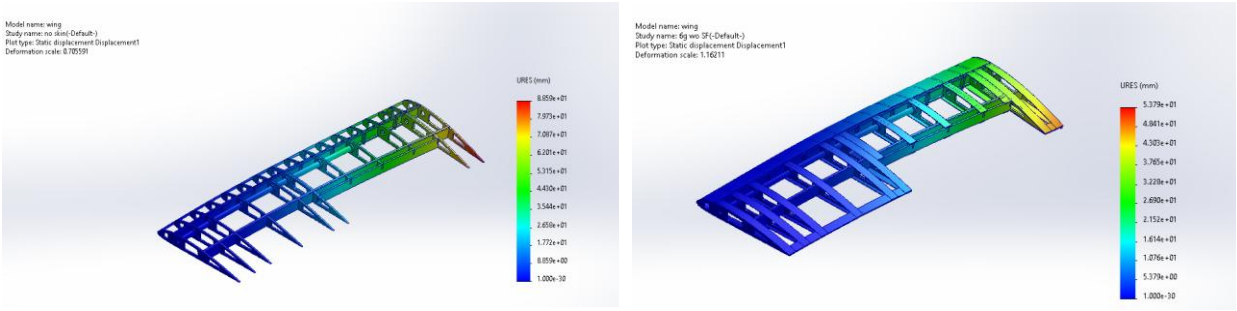


Figure 5.2.1d FEA of Wing with and without Sheeting under 7.38 lb-ft Torque

5.2.2 Fuselage Structure

The fuselage was designed to accommodate three passenger ducks, one hockey puck, and all avionics components while enabling rapid loading and unloading for efficient ground mission performance. Internal layout was driven by passenger and cargo requirements, with two key constraints: wiring could not pass through the passenger compartment, and the center of gravity needed to be positioned ahead of the aerodynamic center for stability. As a result, the heavier cargo compartment was placed forward of the passenger compartment, with the avionics bay located furthest forward.

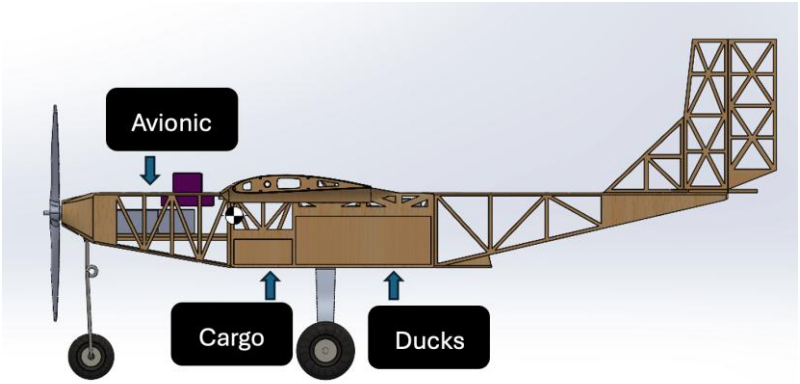


Figure 5.2.2a Fuselage Side View Indicating the Main Compartments

The hockey puck's flat-up orientation set the minimum fuselage width at 3.63 in to minimize drag, accounting for the 3.00 in puck diameter, two 0.118 inside truss panels, and two 0.197 in cargo holder panels. The cargo box bonded to the truss panels served as both payload containment and primary load transfer structure, resisting shear forces from the wing. The passenger compartment used the same width for manufacturing simplicity, with extra space accommodating the restraint system.

To distribute aerodynamic loads efficiently, formers made of 0.197 in balsa-plywood sandwich with circular edges were placed at key locations like the wing seat. Reinforced with longerons and stringers, they formed a semi-monocoque structure resisting bending and torsion while providing a streamlined profile. This approach minimized weight, ensured stiffness against aeroelastic effects, and supported high-speed flight. A teardrop-shaped cowling further reduced parasite drag.

To assess the fuselage's capability to efficiently transfer loads from the wing, tail, avionics, and payload, a finite element analysis was performed to evaluate structural integrity under critical loading conditions. The model was subjected to a 5g loading condition at a required high-speed configuration with a 1.75 safety factor at maximum payload. Results indicated a peak stress of 965 psi in the longerons just forward of the wing mount, with a maximum deformation of 0.0258 inches at the aft end of the fuselage. Considering Balsa's ultimate strength of approximately 2900 psi, the design provides adequate strength for the 5g load case with the applied safety factor.

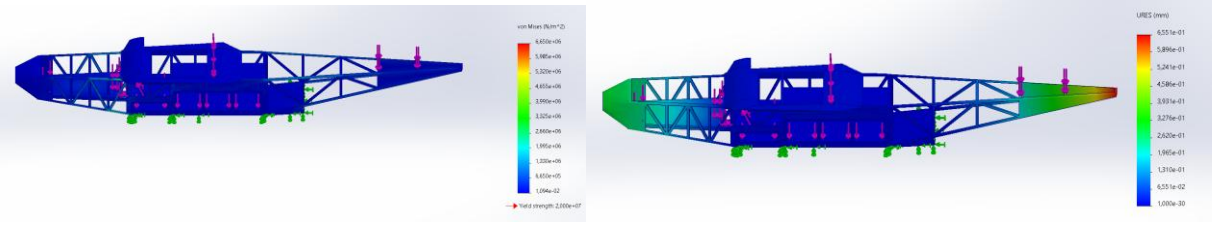


Figure 5.2.2b FEA of Fuselage under 5g MTOW Load

5.2.3 Wing Seat Structure

To minimize aeroelastic effects and flutter, the wing seat was designed as a separate solid structure permanently fixed to the fuselage, unlike Prototype 1 where it was integrated with the longerons. Lessons from Prototype 1 showed significant flutter when wings were installed as separate halves. In this design, the wing seat was 3D printed as a single unit. Two holes at the front accept alignment tubes. The wing halves were joined first, with carbon fiber tubes at the leading edge roots inserted into corresponding slots in the wing seat. Screws provided additional retention. This approach addresses flutter caused by manufacturing errors that compromised wing symmetry in Prototype 1. A faired surface at the wing-fuselage intersection reduces interference drag. The design increases wing rigidity, minimizes aeroelastic effects, and supports high-speed flight objectives.

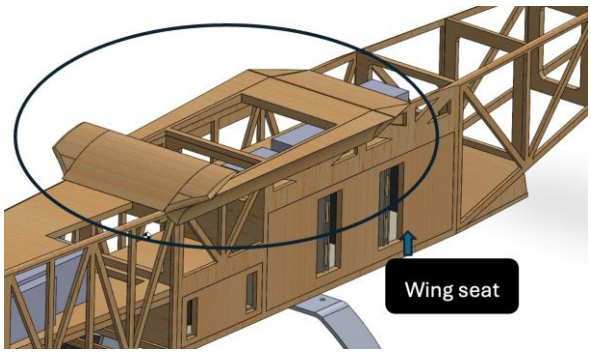


Figure 5.2.3 Wing Seat of Prototype 2

5.2.4 Cargo Compartment Structure

As established, carrying one hockey puck for Mission 2 led to a simple cargo holder box focused on weight minimization and ground mission efficiency. The box featured one open side and was positioned at the fuselage bottom with internal dimensions of 1.4 in height, 3.0 in width, and 3.4 in length. It was intentionally

oversized to accommodate puck size variations and simplify loading, as a tighter fit would hinder handling and increase ground mission time. Foam padding was applied to interior surfaces to prevent puck movement during flight despite the loose fit. A latch with a pusher foam block on its inner face served as the securing mechanism; when closed, the foam forced the puck against the opposite wall, locking it in place. A simple door lock with tape hinges secured the latch, converting the clearance fit into an interference fit for flight. Inspection windows cut into the latch allowed visual verification of proper retention without interfering with the foam block.



Figure 5.2.4a Structure of Payload Compartment in Prototype 2

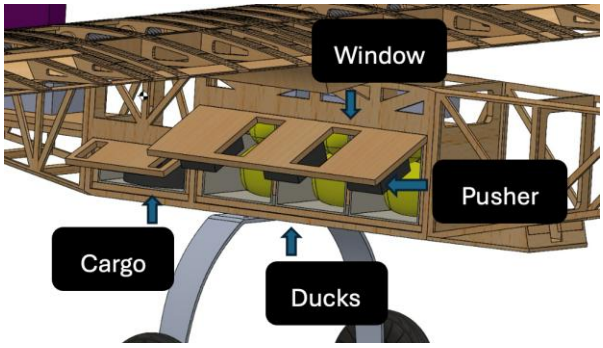


Figure 5.2.4b Structure of Passenger Compartment and Payload Compartment

5.2.5 Passenger Compartment Structure

As established, carrying three passengers for Mission 2 led to a simple passenger holder box with a retaining mechanism similar to the cargo compartment. The box was oversized with foam padding to accommodate duck size variations. A closing latch featuring inspection cutouts and pusher foam forced the ducks against the side wall upon closure, securing them with no remaining degrees of freedom once locked. Unlike the cargo compartment, wooden strip separators were added between each duck station to prevent longitudinal translation, as irregular duck shapes would otherwise allow shifting even with the latch secured. These strips provided individual resting surfaces, with additional foam applied around them to accommodate size variations while maintaining secure retention.

5.2.6 Tail Structure

The empennage plays a critical role in maintaining the aircraft's flight envelope, manoeuvrability, and stability. Although the tail assembly is not the primary structural component of the aircraft, FEA is desired to evaluate potential enhancements and identify reinforcement strategies for the current design.

The static analysis applies a uniform distributed load on the horizontal stabilizer to represent a 2.5g tail lift condition, which typically accounts for approximately 10–20% of the total aircraft lift. For the $Sh = 116.66\ in^2$ (approximately $0.075\ m^2$), assuming the tail contributes 10–15% of total lift, a conservative total load of 10 N is applied (distributed evenly across the surface, directed upward or downward depending on the positive or negative g case) to simulate the aerodynamic force experienced in flight. For the vertical stabilizer, a side load of 5 N is applied to replicate yaw torque induced by banner towing or crosswinds. These load cases enable detailed examination of stress distribution and deformation in the truss structure, mortise-and-tenon joints, and epoxy-bonded interfaces, facilitating identification of optimal reinforcement methods, such as additional carbon fibre doublers and adjustments, to improve durability and stability during Mission 3 operations.

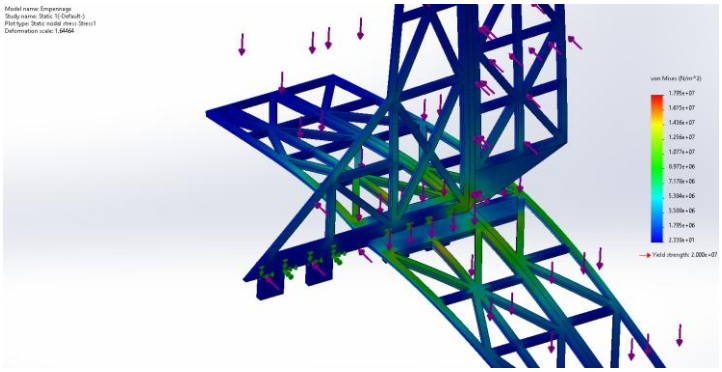


Figure 5.2.6 Von Mises Stress at Maximum UDL

5.2.7 Landing Gear Structure

As established in Section 3.4.5, the tricycle landing gear configuration was selected not only for its stability during take-off, landing, and ground handling but also for its simpler integration of the banner mechanism integration beneath the fuselage. The main landing gear, with a mass of 0.38 lbs., utilizes a commercially bought 6061-T6 aluminum alloy strut featuring curved fillets to mitigate stress concentrations. The 0.12-inch-thick strut, with 0.24-inch diameter axle holes, was bandsaw-cut and manually bent to achieve a propeller clearance of 2 inches and a fuselage clearance of 6.26 inches. This generous initial ground clearance was necessitated by two factors: accommodating a 3.54-inch-tall banner mechanism prototype beneath the fuselage and preventing tail strike during take-off at angles of attack up to 13 degrees. The strut tapers from a mounting face width of 1.2 inches at the fuselage interface to 0.85 inches at the wheel

axle. Tread width was determined to be approximately 12.3 inches based on a stability-based empirical calculation [12].

In previous prototypes, the main landing gear was positioned to maintain a static margin on ground of 15%, corresponding to 85% of the aircraft's static load on the main gear and 15% on the nose gear [9]. However, the integration of a banner mechanism nearly spanning the entire length of the aircraft induced a significant CG shift. To compensate for this added weight and pitching moment, the main landing gear was repositioned to achieve a static margin on ground of 36%, resulting in 64% of the aircraft weight being supported by the main gear. This configuration ensures the aircraft stands firmly in both payload-loaded and payload-unloaded states while still directing most static and landing impact loads through the main landing gear. An FEA analysis was conducted on the above design to validate the structural integrity of the main landing gear under maximum expected dynamic loading. In this simulation, the worst-case load scenario was assumed to occur during landing, with a 5g load factor applied to the static load carried by the main gear (64% of the aircraft's 5.95 lbs mass, excluding landing gear weight). This corresponds to the Mission 2 configuration with the highest anticipated landing loads. An additional safety factor of 1.5 was applied, yielding an ultimate test load of 115.69 N. As shown in Figure 5.2.7a, the resulting von Mises stress distribution remains below the yield strength of 6061-T6 aluminum, confirming that the proposed design possesses adequate strength margins for the intended operational requirements.

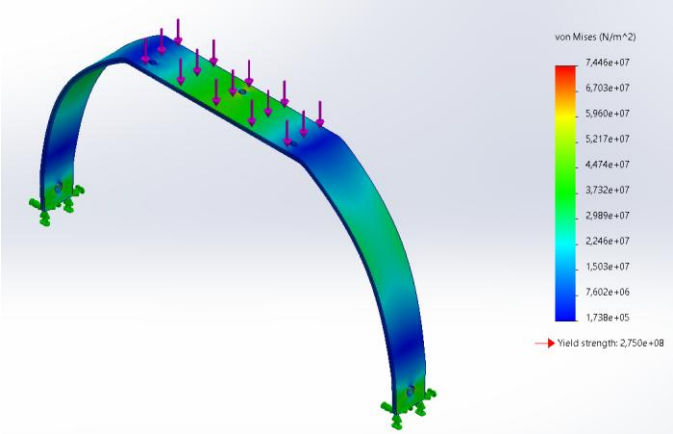


Figure 5.2.7a FEA showing Von Mises Stress at Maximum Loading

The nose landing gear comprises a lightweight 0.12-inch diameter stainless steel rod that extends into the nose of the fuselage, connected to a servo linkage via a metal pushrod. An integrated coil spring embedded within the rod acts as the primary shock absorption mechanism. This design was selected following a trade study which determined that more complex mechanical struts, such as spring-damper arms or oleo-pneumatic struts, introduced excessive weight, cost, and manufacturing complexity for marginal gains in damping performance relative to the lightweight coil spring configuration. The nose gear also adopts a

positive rake angle of around 7 degrees which helps suppress shimmy oscillations as well as creates a built-in-centering caster effect for improved directional stability.

The nose gear wheels are fabricated from polyurethane molded foam, chosen for its low density and adequate energy absorption. The wheels are intentionally narrow to minimize frontal area and associated drag. Overall nose gear height was geometrically determined to maintain the fuselage and propeller clearances established by the main landing gear geometry, ensuring consistent ground contact across all three-wheel attachment points.



Figure 5.2.7b Nose Gear CAD Model

5.3 Banner Mechanism Structure

To maintain rigidity and ensure the successful deployment and release of the banner, a newly designed mechanism as shown in Figure 5.3a consists of two grippers for holding the banner during stow and tow stages which were controlled by servos for banner deployment. A badge reel was stored behind the passenger compartment for banner release as shown in Figure 5.3b. This prevented soft wires from being stuck on the hooks on the aluminum tube, and ensured the banner was securely fastened.

The 3D printed gear-driven gripper mechanism and the servo, fixed by a 3D-printed adjustable fixture were attached to the two ends of a rectangular aluminum tube. All three components are connected using M2 Pan Cross Head screws and an M2 hex nut. A crawling placed in front of and behind the front and back servo to improve the aerodynamic performance of the banner mechanism. The tube went through three wooden U-shape grooves which connected a 0.04in plywood board. Two wooden slots fixed underneath the fuselage allowed the board to go inside, locked by foam and magic tape. The left edge of the banner is fixed by a 3.28ft carbon fibre rod with a payload hanging at the bottom of the rod. The team folded the banner repeatedly in half as a folding method for the ground mission.

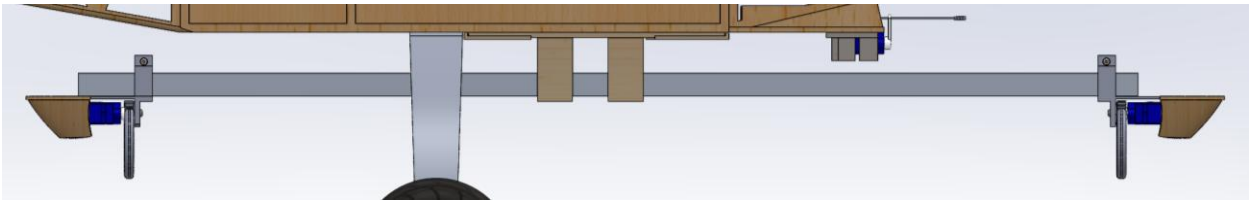


Figure 5.3a Gripper-Mechanism

Four badge reels with a length of 1.97ft and a wire diameter of 0.031in were connected one by one to extend the wire to 5.91ft long, where the end of the wire was tied to a two-sided badge reel attached to the banner. The badge reel was placed inside the fuselage and fixed by a 3-sided triangular wooden slot. The backside of the slot with a length of 3in was blocked by the servo arm of an arm fixed underneath the slot. The hole at the centre of the backwall of the slot allowed the wires from badge reels to go out when pulled by the banner. When the servo rotated 90 degrees, the backwall dropped which released the badge reels and the banner. An FEA analysis of the banner mechanism was conducted on SolidWorks which proves the rigidity of the gripper mechanism as shown in Figure 5.3c, 5.3d and 5.3e.

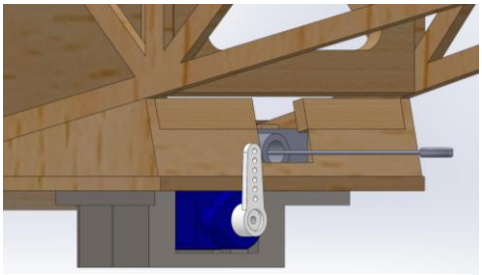


Figure 5.3b Badge Reel Release

From the FEA result, all deformations acting on the gripper mechanism, the stow mechanism, and the badge reel were smaller than 0.0039in. It was determined that this mechanism was sufficient for the rigid and steady motion needed for banner deployment and release.

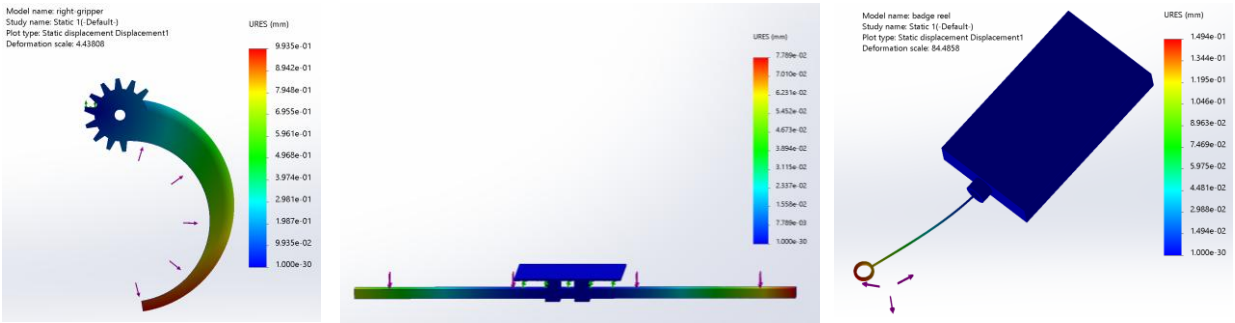


Figure 5.3c) FEA on Right Gripper, 5.3d) Banner Stow Mechanism, 5.3e) Badge Reel FEA

5.4 Avionics and Wiring

The mass of GRIFFIN'S avionics and the components' installation locations were shown in Table 5.4 and Figure 5.4 respectively. To reduce drag due to the large frontal area of the plane, the compartment located in front of the wing was compactly designed. The motor was mounted at the front of the firewall, which was located inside the nose cover. The battery and the ESC were strapped together right behind the firewall, while a fuse holder with an 80A blade fuse installed was strapped to the side wall. A GAONENG 4500mAh 6S 70C Li-Po battery was selected due to its low internal resistance, high C-rating and lightweight compared to similar

Avionics Components	Mass(lbs)
EOLO 13x7 Propeller	0.057
SunnySky X3120 V3 KV 585 motor	0.407
80A Hobbywing Skywalker UBEC (ESC)	0.191
Frsky Archer Plus Sr12+ Receiver	0.051
80A Maxi Blade Fuse	0.137
Fuse holder	0.201
GAONENG 4500mAh 6S 70C Propulsion Battery	0.903
ATECH Infinity 550mAh 2S 85C Receiver Battery	0.084
Hobbywing UBEC 10A-Car	0.075
KM9040MD Servo(s)	0.220
Guohua A0090 Servos (banner deploy & release)	0.06
Total	2.39

Table 5.4 Avionics Components Masses

COTS options. An 80A Hobbywing Skywalker UBEC (ESC) was selected due to its compact size. A UBEC, receiver battery and receiver were placed under the wing. An external switch was placed on the side wall of the fuselage near the wing's trailing edge to disconnect the receiver from the battery when needed. A Frsky Archer Plus Sr12+ Receiver was selected as it supports up to 12 channels for control and telemetry. Guohua A0090 servos were chosen specifically for banner deploy and release mechanisms due to its low weight, high torque performance.

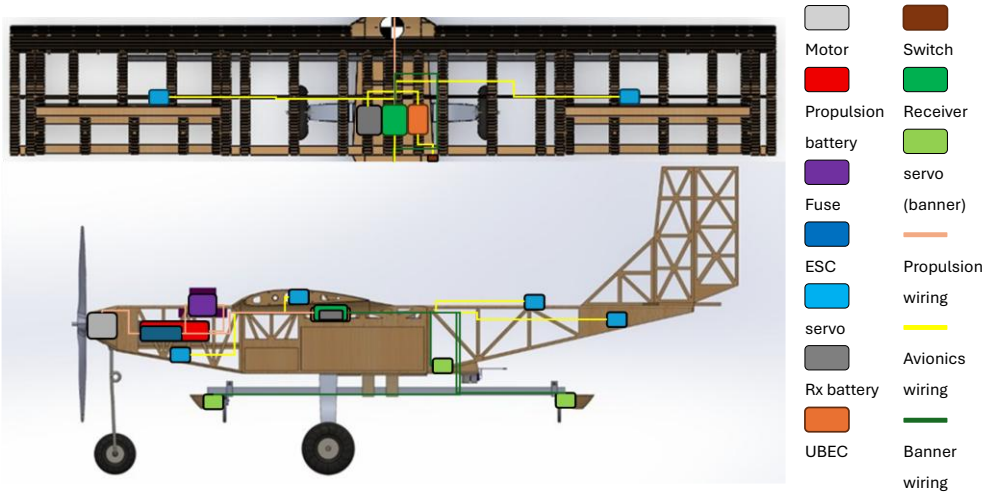


Figure 5.4 GRIFFIN Avionics and Wiring Diagram

5.5 Weight and Balance

A proper center of gravity is essential for longitudinal control and stability. All CG analysis was conducted in SolidWorks using models of each component with accurate weights and material densities and subsequently validated using a physical scale. The results for all three missions are presented below. The banner was stowed near the CG during takeoff to reduce the CG shift upon deployment.

Key components of Prototype 2 GRIFFIN are listed in the table below, including their weight and relative position behind the aircraft firewall. The fuselage position is based on the location of its center of gravity. The motor, motor mount, and propeller have negative positions as they are located forward of the firewall. Wires, additional electronics, fasteners, adhesives, and other miscellaneous items are distributed throughout the aircraft and therefore do not have assigned positions.

Component	X-Position (in)	Mass (lb)		
		M1	M2	M3
Fuselage	9.04	0.46	0.46	0.46
Main Wing	12	0.32	0.32	0.32
Empennage	37.52	0.08	0.08	0.08
Front Landing Gear	2.03	0.16	0.16	0.16
Main Landing Gear	14.58	0.38	0.38	0.38
Propulsion Battery	3.91	1.10	1.10	1.10

Motor	-1.63	0.39	0.39	0.39
Motor Mount	-0.06	0.01	0.01	0.01
Propeller	2.54	0.06	0.06	0.06
ESC	14.53	0.19	0.19	0.19
Fuse and Fuse Holder	6.93	0.34	0.34	0.34
Receiver	15.10	0.05	0.05	0.05
Receiver Battery	15.10	0.08	0.08	0.08
UBEC	15.10	0.07	0.07	0.07
Wires and Additional Electronics	N.A.	0.20	0.20	0.20
Adhesives and Others	N.A.	0.50	0.50	0.50
Ducks	15.84	N.A.	0.13	N.A.
Hockey Pucks	10.41	N.A.	0.375	N.A.
Banner and Connection Tools	16.60	N.A.	N.A.	0.88
Aluminum Tube For Banner	16.60	N.A.	N.A.	0.09
Total Weight		4.39	4.90	5.36
CG Location		9.04	9.08	9.10

Table 5.5 Weight and Balance Table

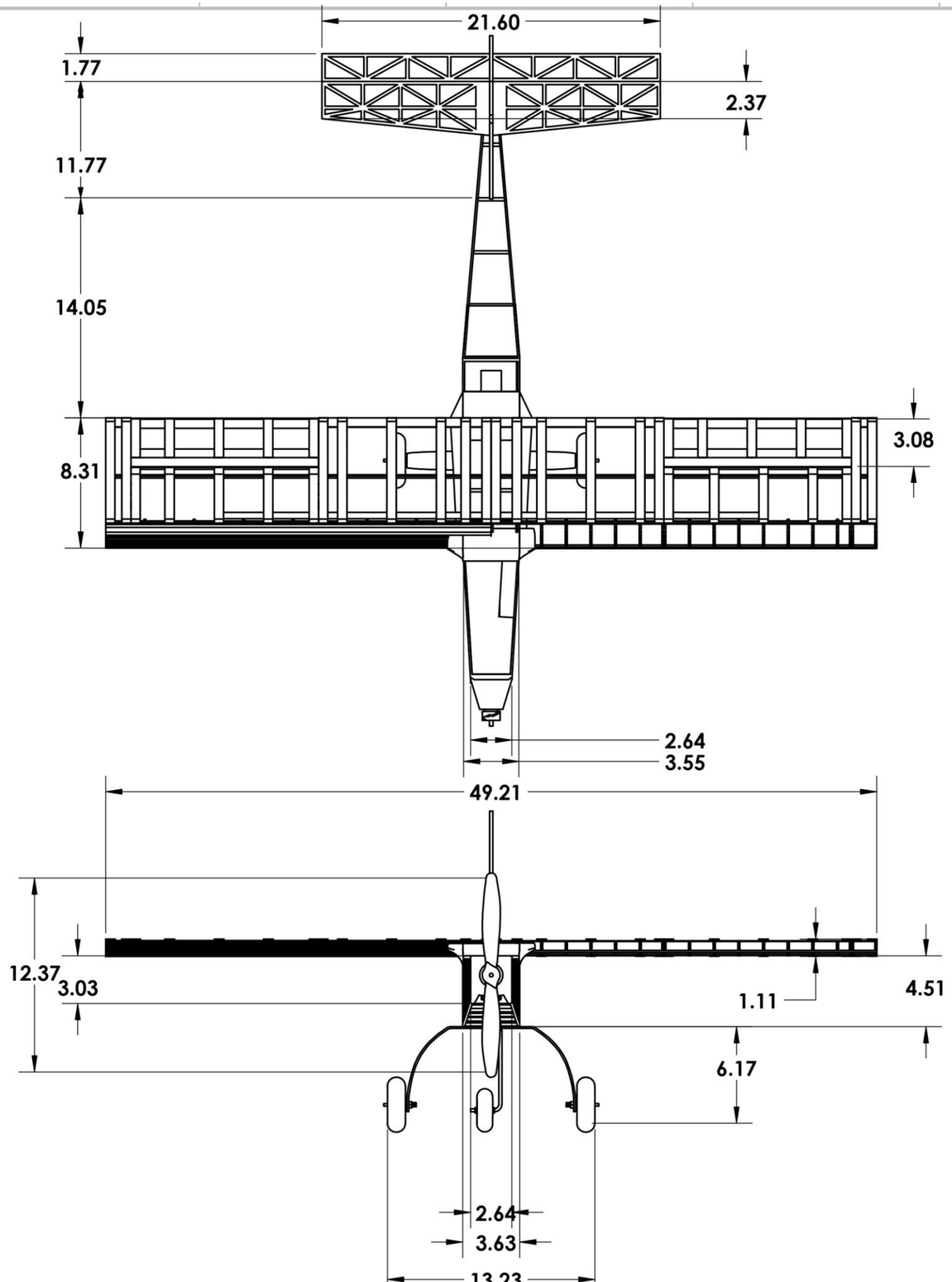
5.6 Flight Performance & Mission Specific Performance

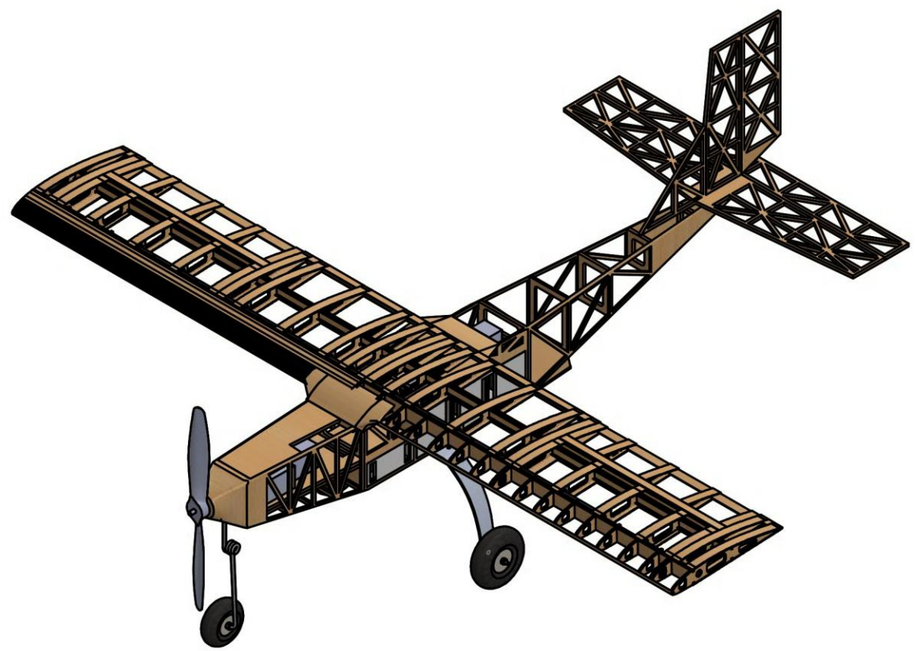
Utilising the detail design of the aircraft, the flight performance and the mission specific performance were analyzed and resulted in the table 5.6 below.

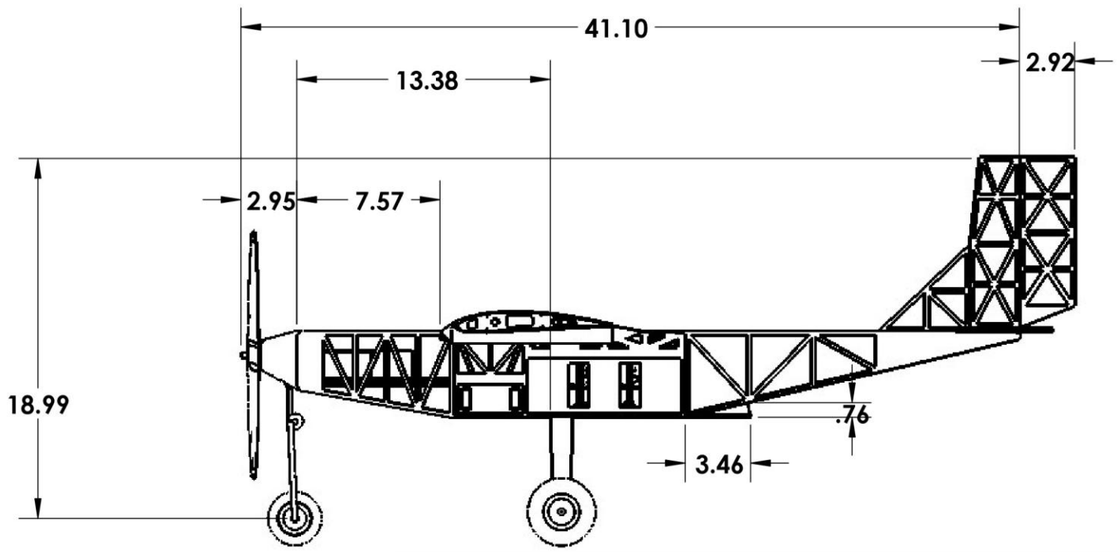
Parameters	GM	M1	M2	M3
Payload (lbs)	N/A	N/A	0.509	N/A
TOW (lbs)	N/A	5.29	5.8	5.95
TOFL (ft)	N/A	65	65	65
Average Throttle (%)	N/A	80	80	80
Average Speed (ft/s)	N/A	104.98	100	59
Cruise Speed (ft/s)	N/A	105.0	105.0	55.9
Maximum Speed (ft/s)	N/A	134.5	132	68.9
Turning Speed (ft/s)	N/A	104.98	100	55
Battery Usage (WAh)	N/A	62	62	62
Number of Laps	N/A	3	9	5
Mission Time (sec)	25	93.75	281.25	293

Table 5.6 Flight Performance Parameter Table

5.7 Drawing Package

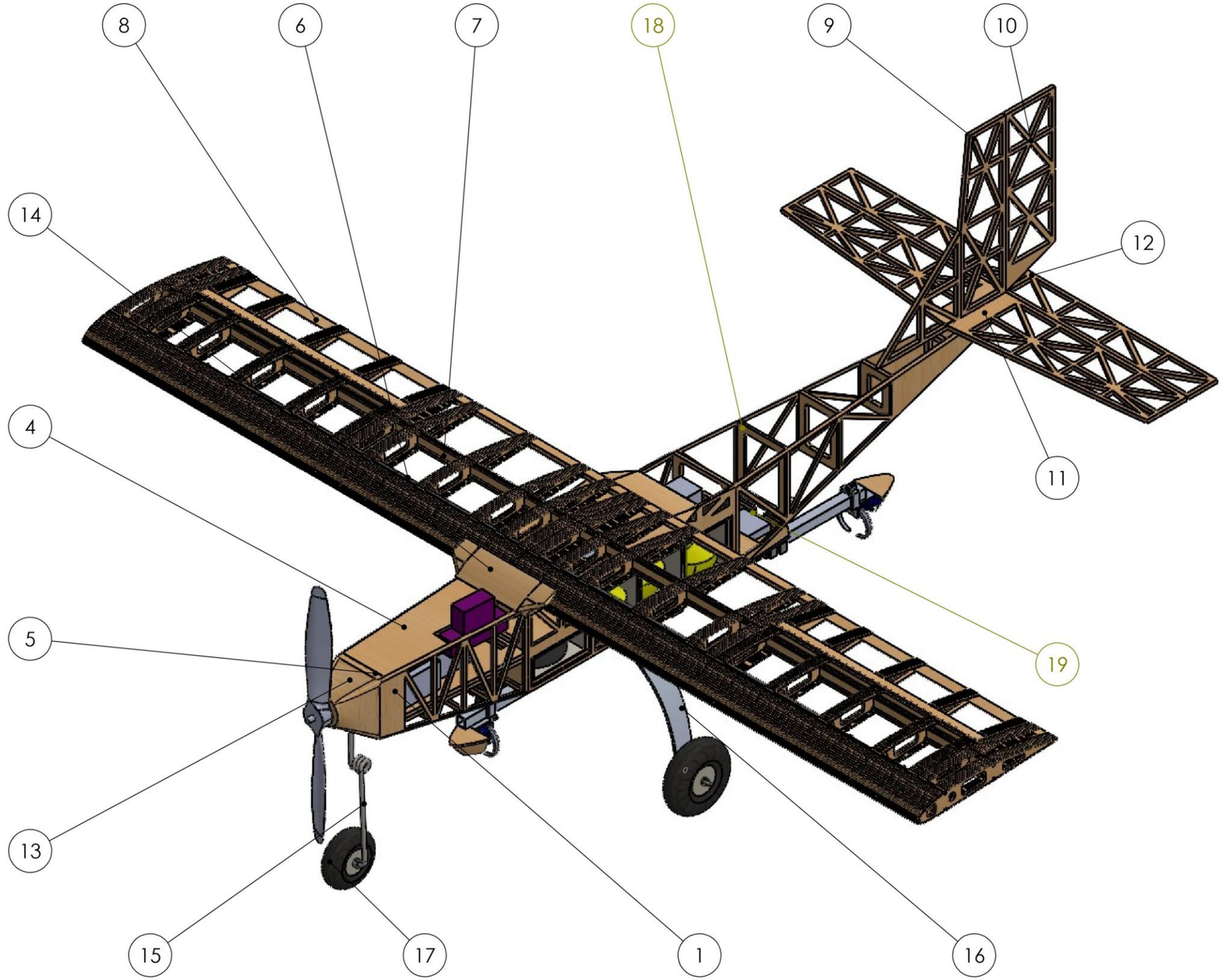






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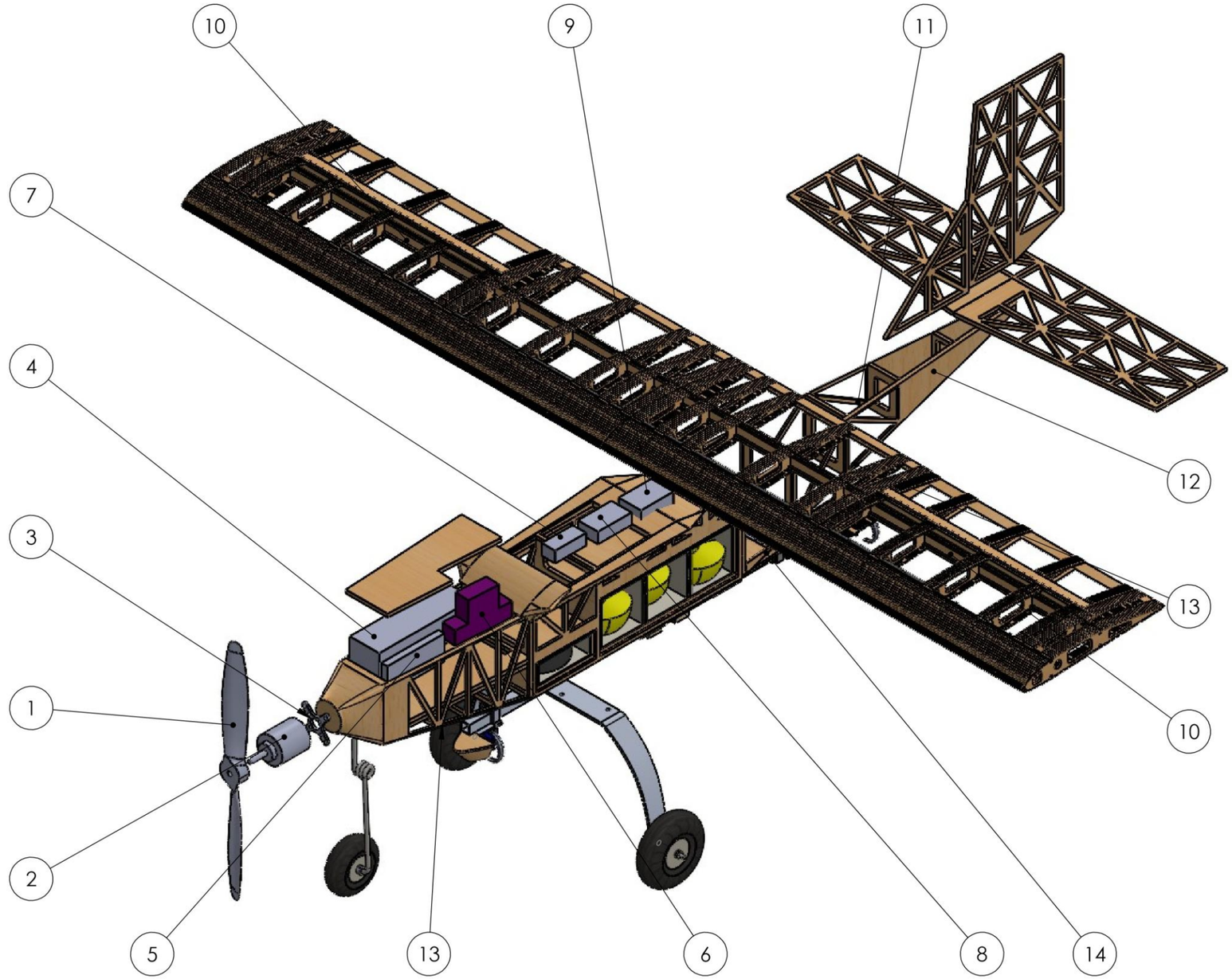
ITEM NO.	PART NUMBER	MATERIAL	QTY.
1	FUSELAGE	BALSA AND PLYWOOD	1
2	CARGO LATCH	PLYWOOD AND PE FOAM	1
3	PASSENGER LATCH	PLYWOOD AND PE FOAM	1
4	AVIONICS LATCH	BALSA	1
5	FIREWALL	BALSA AND PLYWOOD	1
6	FRONT WING SPAR	CARBON FIBER	1
7	AFT WING SPAR	BALSA	1
8	AILERON	BALSA	2
9	VERTICAL STABILIZER	BALSA	1
10	RUDDER	BALSA	1
11	HORIZONTAL STABILIZER	BALSA	1
12	ELEVATOR	BALSA	1
13	NOSE COWLING	PLA	1
14	WINGSEAT	PLA AND BALSA	1
15	FRONT LANDING GEAR	6160-T6 ALUMINIUM	1
16	MAIN LANDING GEAR	6160-T6 ALUMINIUM	1
17	WHEELS	POLYURETHANE	3
18	FUSELAGE BULKHEADS	BALSA AND PLYWOOD	7
19	BANNER RELEASE HOLDER	BALSA AND PLYWOOD	1



COVERING FILM IS OMITTED FOR CLARITY PURPOSE

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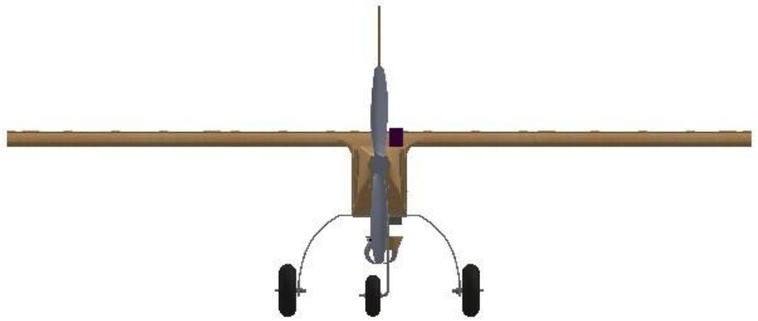
ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	PROPELLER	EOLO 13x7	1
2	MOTOR	SUNNYSKY X3120 V3 KV 585 MOTOR	1
3	MOTOR MOUNT	SUNNYSKY X3120 V3 KV 585 MOTOR	1
4	PROPULSION BATTERY	GAONENG 4500MAH 6S 70C	1
5	ESC	80A HOBBYWING SKYWALKER UBEC (ESC)	1
6	FUSE AND FUSE HOLDER	80A MAXI BLADE FUSE	1
7	AVIONICS BATTERY	ATECH INFINITY 550MAH 2S 85C AVIONICS BATTERY	1
8	RECEIVER	FRSKY ARCHER PLUS SR12+ RECEiVER	1
9	UBEC	HOBBYWING UBEC 10A-CAR	1
10	AILERON SERVO	KM9040MD SERVO	2
11	RUDDER SERVO	KM9040MD SERVO	1
12	ELEVATOR SERVO	KM9040MD SERVO	1
13	BANNER RELEASE SERVO	GUOHUA A0090 SERVO	2
14	BANNER DEPLOY SERVO	GUOHUA A0090 SERVO	1

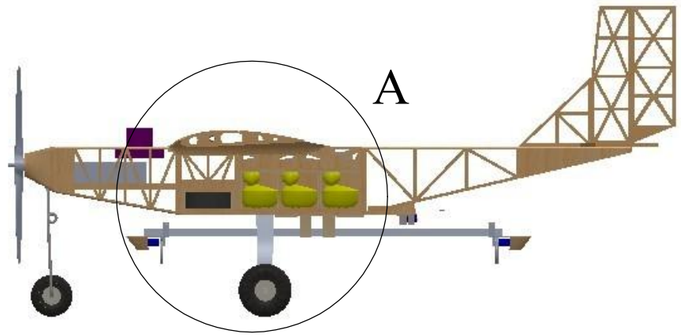


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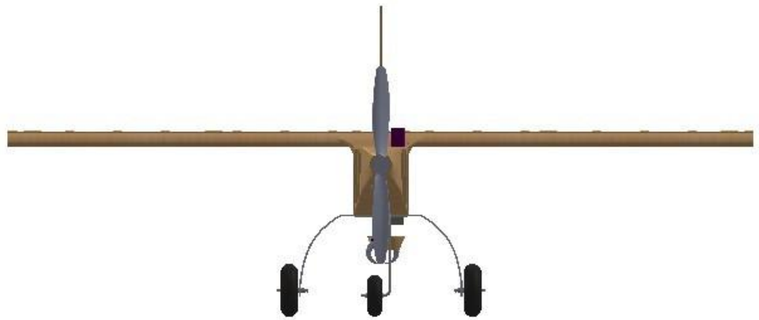
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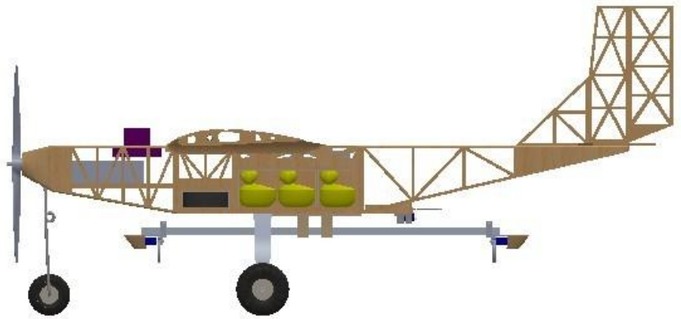
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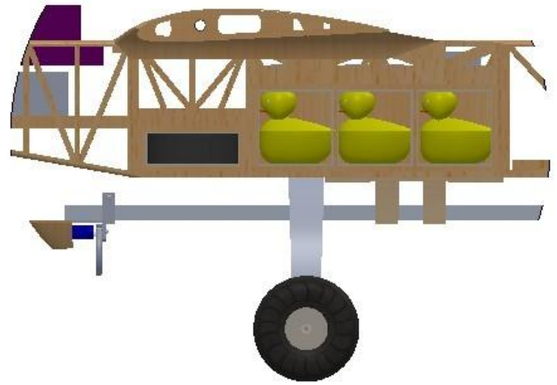


MISSION 3 CONFIGURATION



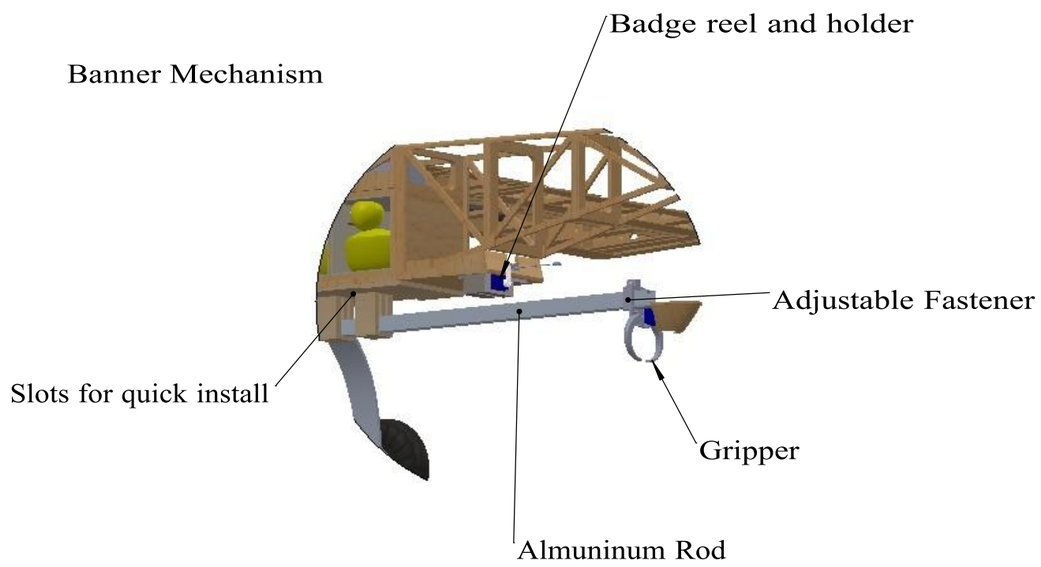


Passenger and cargo placement



DETAIL A

SCALE 1 : 5



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Drawn by: Srazeen Gurung	The Griffin	
Checked by: Zhao Zehui	Size B	Mission Layout
	All Dimension in Inches	Sheet 4 of 4

6 Manufacturing Plan

The team evaluated numerous manufacturing processes and materials for each major aircraft component to identify the optimal combination of weight, ease of repair, manufacturing speed, team experience, and cost. The selected fabrication methods for each component are detailed in the following subsections.

6.1 Manufacturing Processes Investigation

6.1.1. Built-Up Balsa and Plywood

The team used a laser cutter to cut through low-density balsa wood boards according to finalized CAD models. Balsa was chosen for its high strength-to-weight ratio and easy repairability during the prototyping stages of the manufacturing process. The resulting balsa wood parts were glued together using cyanoacrylate (CA) adhesive to form the precise balsa truss structure of the fuselage with lightening holes, the wing ribs and spar webs, the dihedral braces of the wing, and the star-like pattern of the empennage.

For critical structural components such as the landing gear mount plate and the fuselage formers/bulkhead, aircraft-grade plywood was used to withstand the concentrated loads, the impact/vibration, and the shear/torque generated by bolts and screws these components experience. For the firewall and the nose of the plane, a sandwich composite of balsa reinforced on either side by two pieces of plywood was considered for maximum strength for the propellor and landing impact forces experienced by the two parts respectively.

6.1.2. Aluminum and Stainless Steel

For load-bearing components in the model aircraft, 6061-T6 aluminum and stainless steel provides favorable strength, manufacturability, and reliability. 6061-T6 aluminum was the preferred choice for primary structures such as the main landing gear strut due to its excellent strength-to-weight ratio, cost and availability, and ease of fabrication through waterjet cutting, bandsaw machining, and metal bending. Stainless steel, although heavier, was selectively employed for high-stress, small-volume components like axles, control linkages, and the nose landing gear strut where its superior hardness, wear resistance, and ability to withstand concentrated point loads without deformation are critical.

6.1.3. 3D Printing

Professional grade 3D printers were mainly used for the rapid prototyping of complex shapes that would otherwise be difficult to manufacture through conventional additive manufacturing techniques, such as an aerodynamic propellor cover for the fuselage nose, prototype compartments for the passenger and cargo, and fairings for the main landing gear wheels

6.1.4. Reinforcements

Carbon fiber strips bonded with CA glue and epoxy reinforced high-stress regions such as wing joints and fuselage truss intersections. Chopped carbon fiber "hair", mixed with epoxy, created lightweight

reinforcements over abnormal shapes such as the nose gear mounting. Triangular wooden gussets were installed at all major structural intersections to resist racking and torsion. Fiberglass tape laminated with epoxy provided additional shear strength over seams and joints. Finally, a heat-shrink film was used to cover the entire airframe, delivering a smooth, aerodynamic surface that encases all avionics and compartments.

6.2 Manufacturing Processes Selection

The manufacturing process is tightly integrated with design and simulation to ensure performance and reliability. The manufacturing plan's strength lies in its logical progression and the integration of both additive and subtractive manufacturing techniques, tailored to the specific requirements of each subsystem. This linear flowchart is driven by a series of critical iterative loops essential for mitigating risk and optimising team performance.

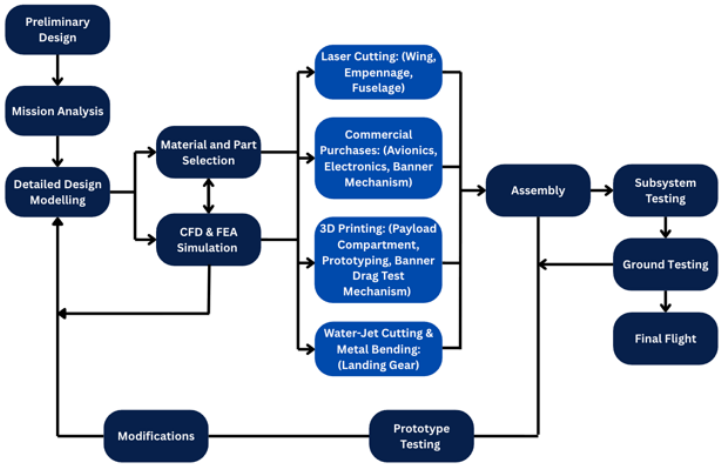


Figure 6.2 Manufacturing Plan

6.2.1. Fuselage

The fuselage had all truss members designed in AutoCAD and precision laser-cut from 0.118-in balsa wood. Each component featured integrated notches that allowed the pieces to interlock like a puzzle, ensuring accurate alignment during assembly. For the side panels, a 1:1 scale template of the fuselage profile was printed and secured to a work surface with box tape. Each balsa member was positioned and glued on this template, enabling precise, repeatable, and simple tool-free assembly. Laser-cut 0.197-inch balsa formers with rounded corners were glued to the side panels, integrating with the pre-assembled passenger and cargo boxes. Metal square blocks maintained precise parallel alignment during curing to ensure consistent fuselage width. Wooden stringers were then applied along the former edges to provide a mounting surface for the heat-shrink film covering. The wing seat was 3D-printed and attached to the fuselage using a two-stage bonding process: CA glue for initial positioning, followed by structural epoxy to create a permanent bond.

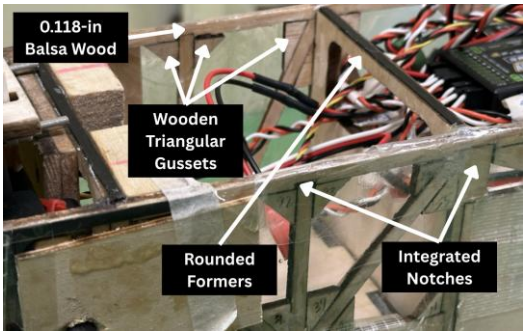


Figure 6.2.1 Fuselage Manufacturing Components

6.2.2. Wing

The wing was fabricated from laser-cut 0.12-inch balsa ribs, spars, webbing, and reinforcement rings exported from DXF files. Assembly followed a jig-based method: rib combs were positioned over a printed planform, ribs were then seated into the combs, and the front upper spar was installed to establish alignment before gluing. Shear webs were glued between spars to complete the primary box structure. The leading edge was sanded to the final airfoil contour, and the D-box was formed by gluing pre-moistened 0.04-inch balsa sheeting over the forward section. I-beam reinforcement was achieved by bonding 0.6-inch strips of 0.04-inch balsa along the rib surfaces. The ailerons were assembled separately using balsa strips and attached via Monokote heat-shrink film, which also served as the hinge. The entire wing was covered with Monokote, the two halves were joined with structural adhesive and Monokote, and servo cutouts were made for final integration.

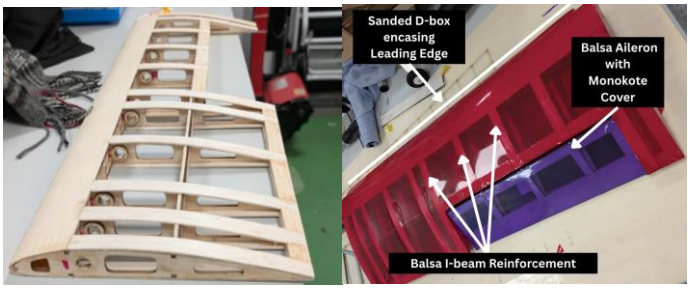


Figure 6.2.2 Wing Manufacturing Components

6.2.3. Empennage

The entire empennage was divided into separate parts and exported as DXF files for laser cutting from 0.197-inch low-density balsa sheets. These parts were assembled using an engraved plywood plate, which served as a jig or reference based on projections of both the horizontal and vertical stabilizers, and then fixed in place with CA glue. Specific high-stress areas identified in section 5.2.6 were reinforced with carbon fibre strips and CA glue to minimize deformation and enhance overall structural integrity.

The attachment method utilizes multiple mortise-and-tenon joints: the tenon of the vertical stabilizer passes through the mortise in the horizontal stabilizer and fuselage tail, extending through a support plate designed

to reduce the degree of freedom. The horizontal stabilizer is permanently bonded to the fuselage using epoxy resin. To ensure perpendicular alignment between the vertical and horizontal stabilizers, a triangular plywood strip is adhered to the junction gap with epoxy, eliminating installation misalignment.

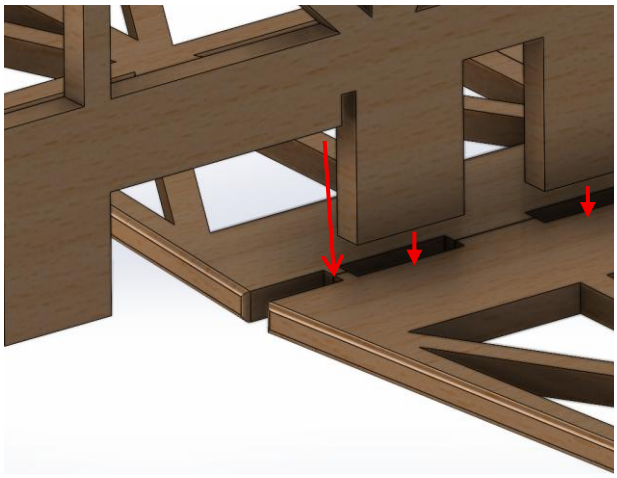


Figure 6.2.3a Empennage's Mortise-and-Tenon Joints

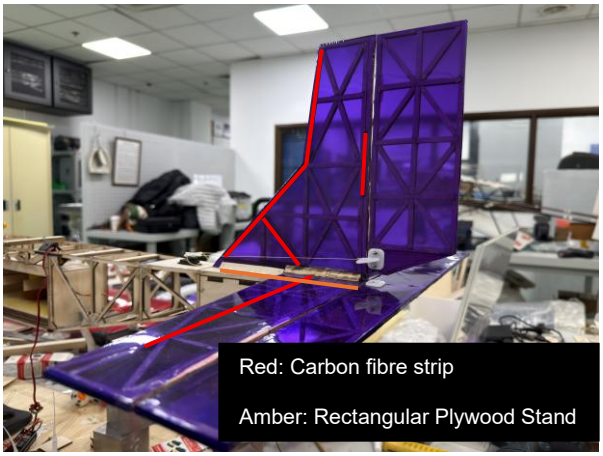


Figure 6.2.3b Empennage's Reinforcement

6.2.4 Landing Gear

The main gear utilizes a commercially sourced 6061-T6 aluminum alloy strut as it is a tried-and-tested metal that is easy to use in subtractive manufacturing processes. Using metal cutting lubricant and a vertical bandsaw machine donning a bi-metal blade, the strut was cut to the desired length and the cut edges were deburred using a manual file to remove any stress-inducing sharpness. 0.24-inch diameter axle mounting holes were drilled at the strut ends with the help of a drill press to prevent work-hardening, and a drill with HSS twist drill bits to round and widen the holes.

The strut was then manually bent using a bench vise with soft jaws to prevent surface marring, with a steel pipe employed as a lever. Bending was performed incrementally, with frequent angle checks using a digital protractor to account for the natural spring back that aluminum metal exhibits when bent. In fact, the strut was bent 2 degrees further than required to account for gradual springback. The final geometry provides a 2-inch propeller clearance, as specified by design constraints. A manufacturing interference issue arose when the attachment bolts, adhering the landing gear to the fuselage bottom, protruded into the passenger compartment. This was resolved by using an additional 0.2-inch balsa wood layer that was laser-cut with recesses to fully embed the protruding nuts, restoring a flush interior surface without compromising structural integrity or passenger space.

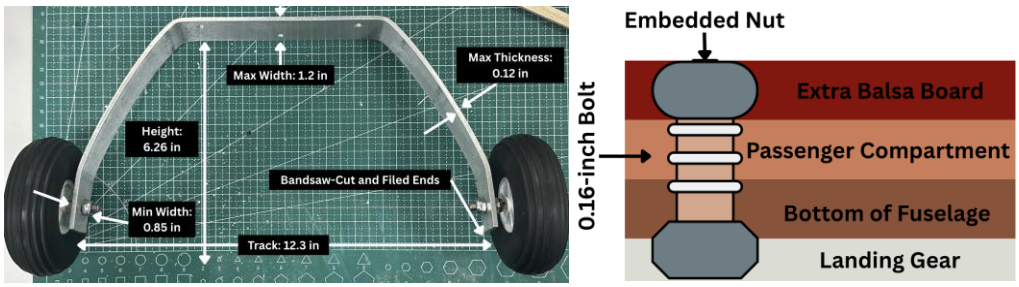


Figure 6.2.4 a) Main Gear Manufacturing Parts b) Solution to Interference Issue

The nose gear comprises a custom-formed coil spring fabricated from 0.12-inch diameter stainless steel rods, cut to length using industrial bolt cutters and an angle grinder. A D-shaped flat was filed onto the rod to accommodate a wheel collar, enabling integration with a nylon servo mount. A rigid metal pushrod links the servo arm to the steering mechanism, the optimal position of which was empirically determined through iterative testing to maximize rotational range of motion.

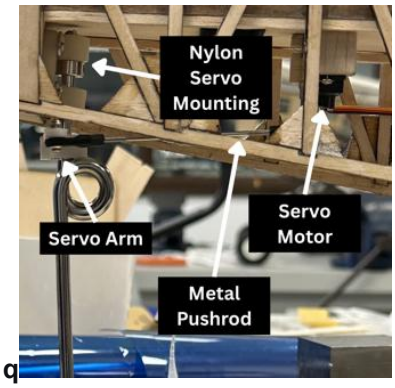


Figure 6.2.4c. Nose Gear Manufacturing Components

6.2.5. Banner Deployment/Detachment Mechanism

The lightweight poly pongee banner, (density: 78g/cm^3) is mounted on a carbon fibre rod with a 3D-printed PLA payload affixed to the rod's lower end to maintain vertical orientation during flight through gravitational restoring moment. The deployment mechanism is attached underneath the fuselage, held by two 3D printed PLA grippers that are actuated by two high-torque servos mounted on a 60cm-long lightweight aluminum rectangular tube. The wires hanging the banner are made of lightweight steel which are retracted in a custom-made wooden badge reel.



Figure 6.2.5c. Banner Mechanism Manufacturing Components

6.2.6. Passenger and Cargo Restraint System

The passenger and cargo restraint system both feature holder boxes and their respective doors with pusher foams attached. The rectangular passenger and cargo holder boxes were constructed using puzzle-like panels laser cut from 0.197-inch balsa wood. This interlocking design allowed the boxes to hold their shape even without adhesive, facilitating alignment during assembly. However, to ensure sufficient rigidity and the secure retention of the payloads, thin CA glue was applied at all joints to permanently bond the panels together. For the doors, they were manufactured via laser cutting with holes incorporated to allow visual inspection of payload securement. Pusher foam blocks were attached to the doors using simple double-sided tape. The assembled doors were then installed on the fuselage using glass fiber tape serving as the hinge, with a spring latch mechanism functioning as the door lock to secure the compartment during flight.

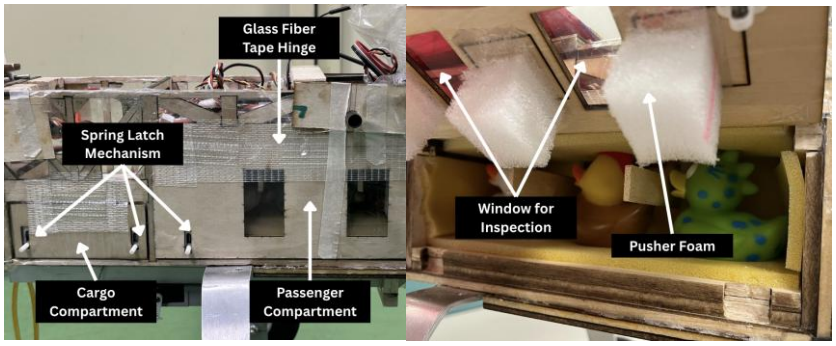


Figure 6.2.6. Passenger and Cargo Compartment Manufacturing Components

6.3 Manufacturing Milestones

A manufacturing Gantt chart was developed prior to prototype #1 to establish a clear schedule and facilitate cross-functional coordination. Progress is tracked by the manufacturing lead to ensure all milestones are met. The schedule shown in Figure 6.3 reflects the production timeline for the two most recent prototypes. Although the 2nd prototype was slightly rushed, the team hopes the template remains consistent across all iterations. This template represents a typical 20-day manufacturing cycle for a single aircraft. This timeline includes allocated time for component testing and redesign, supporting the iterative manufacturing process described in Section 6.2.

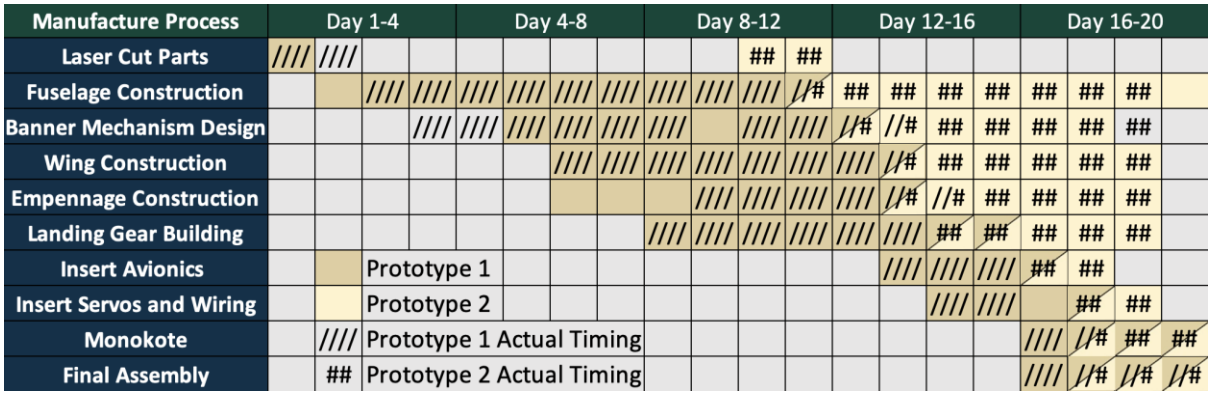


Figure 6.3. Manufacturing Gantt Chart

7 Testing Plan

7.1 Testing Objectives & Schedule

The testing objectives and the schedule are outlined in the following tables.

Division	Method
Avionics	Assess thrust, battery current, power, and RPM across various propeller and motor configurations; obtain static thrust values.
Ground Test	
Structure, CAD	Adding payloads to the restraint systems and record the highest holdable amount.
Aerodynamic, Avionics	Perform the release mechanism of the banner using a transmitter, receiver, and servos.
Aerodynamic, Avionics	Test and conduct actual banner drag test on a prototype with flight controller.
Structure, CAD	Lift the aircraft at its MTOW vertically by the wingtips to verify that it remains level throughout the mission.
Manufacturing, structure, avionics	Simulate each ground mission and log the corresponding times required for completion.
Avionics	Connect all electronic components and validate their functionality.
Aerodynamics	XFLR5 and Ansys to assess the aircraft's stability characteristics and banner drag performance.
Manufacturing, structure	Perform a loading test and inspect the landing gear and associated connections for any damage.
Structure	FEA methods to assess individual components and whole aircraft's structural rigidity.
Flight Test	
All leads	Perform M1, M2, M3 and GM, check the installation of all parts and assembly.

Table 7.1a Testing Objectives

The team conducted one full-day flight test using commercial aircraft to examine performance for the banner mechanism in M3 before the aircraft prototype was manufactured, and two whole-day tests for each of the prototypes. All ground tests mentioned in table 7.1a were conducted every time during each flight test.

Date	Aircraft	Objectives
9th January, 2026	Commercial aircraft	Banner mechanism testing
21st January, 2026	Prototype 1	Testing of M1 & M3 Testing aerodynamic performance and Trimming
4th February, 2026	Prototype 1	Testing of M1 & M3
11th February, 2026	Prototype 2	Testing of M1 & M3 Test aerodynamic performance and Trimming
13th February, 2026	Prototype 2	Testing of M3

Table 7.1b Flight Testing Schedule

7.2 Sub-system Testing

7.2.1 Static Thrust Test

The static thrust test was conducted to obtain power consumption and thrust generated by the propulsion system and refine our design decisions. The setup also followed the competition rule, which consisted of the motor, propeller, propulsion battery, ESC with BEC disabled, and an 80A fuse. This could simulate the actual system running situation (5 minutes mission time) before installing them on the plane. Since the test involves high voltage and high-speed rotation, a safety net and belt barriers were set up to surround the machine to prevent others from approaching it during testing. An arming switch and emergency switch were also connected to the system to prevent the system from being turned on accidentally and stop it in case of emergency.

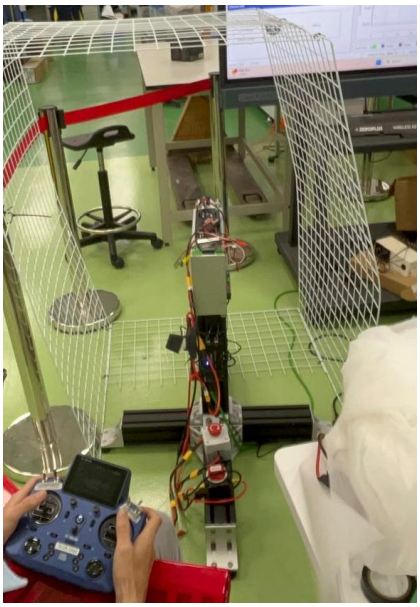


Figure 7.2.1 Static Thrust Test Setup

7.2.2 Ground Test

7.2.2.1 Electronics test

Before the manufacturing process, a standalone control system was tested on a foam board to ensure that all the servos, UBEC, receiver, and receiver battery work according to the radio controller's commands (Figure 7.3.2.1.a). For example, the motion of the two ailerons was set to be opposite to each other and react exponentially, while the rudder and the front landing gear's motion was synchronised. Radio range check test and fail-safe test can also be conducted on the same board. Therefore, when the servos were installed, the team just needed to ensure the servos' wires lie in the same direction as they did on the foam board. This greatly reduced the time required to set up after manufacturing. For radio range check test, the radio controller was put in range check mode, one team member would then walk 30 steps away to simulate the identical real scenario of 900 steps away the plane/board and check the signal quality at low signal strength and another team member would check the control surface movement (Figure 7.3.2.1.b,c). For a fail-safe test, it is conducted by checking whether the control surface movements follow the competition rules after turning the controller off when the receiver is still on. All the control surfaces were checked again before flight tests were conducted at the airfield (Figure 7.3.2.1.c).

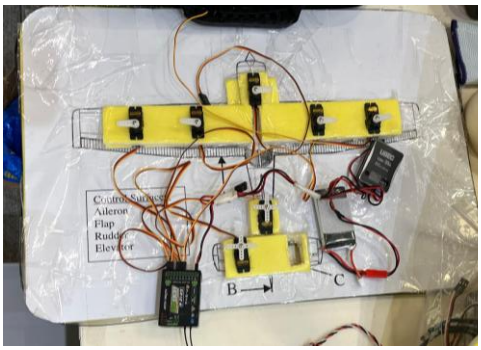


Figure 7.2.2.1.a Electronics test (early stage)



Figure 7.2.2.1.b Radio range test



Figure 7.2.2.1.c Electronics test (airfield)

7.2.2.2. Landing Gear Testing

To ensure the reliability and performance of the landing gear system, the team executed a multi-stage testing campaign. This protocol was designed to quantitatively validate our design against the real-world demands of take-off, landing, and taxiing. The methodology progresses from basic inspections to increasingly demanding structural and functional tests, ensuring the team identifies and designs against any potential failure modes before the fly-off.

Firstly, structural integrity and alignment checks were carried out before every flight. This stage involves a static examination and a full visual inspection of all struts, axles, and mounting structures for any signs of micro-cracks or weak points. Concurrently, the team verified the security of every fastener, D-cut screw, wheel collar, and bolt to ensure no disassembly occurs. Next, using precision tools, the team measured the track width and wheelbase of the aircraft to ensure perfect symmetry and to set the wheel toe and camber angles to zero. This ensures the aircraft will track straight during high-speed take-off and landing rolls, eliminating drift or ground-looping.



Figure 7.2.2.2a Landing Gear Structural Integrity and Alignment Check

Secondly, to simulate a real landing event, the team performed a controlled drop test. Assuming a stopping distance of 1.57 inches, it was determined that the landing gear attached to the fully loaded aircraft had to be dropped from a height of 3.15 inches in order to simulate a 5g landing. This test was repeated multiple times to determine whether the landing gear was functionally refined for smooth and predictable operation in a dynamic flight environment. Lastly, a steering response test was performed on the nose gear connected to a makeshift firewall to verify that it turned freely and accurately responded to control inputs from the servo motor without bottleneck or hesitation.

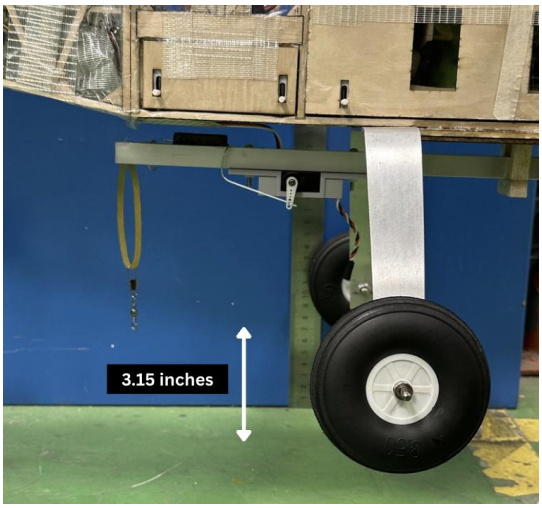




Figure 7.2.2.2b Controlled Drop Test Setup and 7.2.2.2c Steering Response Test Setup

7.2.2.3 Banner Mechanism Deployment Testing

As mentioned in section 5.3, the initial desirable deploy mechanism was the 4-bar linkage bucket. Even though it was not centered underneath the fuselage, it would shift the cg of the aircraft towards one side shown in Figure 7.3.2.3a. In the rubber band mechanism, the team discovered that the plastic wires were trapped by the hooks on the metal rod, which prevented the banner from deploying as shown in Figure 7.3.2.b. In the cage design, the mechanism was not rigid enough to hold the banner as shown in Figure 7.3.2.c. For the spool release mechanism, the gap between the pin and the wooden blocks were hard to lessen to prevent the spool from vibrating vigorously. The metal pin was too soft ()to be pulled out safely from the blocks. The grippers mechanism was determined to be the final design of the banner deployment mechanism.



Figure 7.2.2.3a) Bucket, b) Rubber-band, c) Cage

The team also focused on the impact induced by the drag of the banner during flight and compared different types of materials for a fixture rod attached to the edge of the banner. As shown in Figure 7.2.2.3b, the banner was deployed in the air during the flight.



Figure 7.2.2.3b Banner Deployment

7.3 Flight Test

Several flight tests were conducted to verify the performance of Griffin. Different telemetry sensors, such as GPS, altitude, and airspeed sensors, were used to record the data during most of the flight for both prototypes. For GPS sensors, it recorded the flight route (Figure 7.3.2), which can be imported to Google Earth. We can measure and compare its diameter to the actual diameter (1000ft). By the following equation, we can refine our prediction of the number of laps the plane can fly:

$$laps_{predicted} = laps_{observed} \times \frac{diameter(ft)}{1000ft} \times \frac{5\ mins}{t_{flight\ test}(min)}$$

Sensors	Usages
Frsky ASS70 ADV Airspeed sensor	Monitor airspeed, altitude, ground speed Perform aerodynamic analysis
Frsky GPS ADV sensor Radiolink GPS TS100	
Built-in accelerometer in receivers	Obtain the g-forces act on the wing and the landing gear
Frsky FLVS ADV Voltage Sensors	Check the battery usage and warn the pilot when the battery voltage is low

Table 7.3 Sensors Used in Flight Test and their Usages

As required in AIAA 2025-2026 DBF rules, GPS was only for data collection during flight test for design optimization and would not be used during the mission attempts in 2025 – 2026 competition in US.

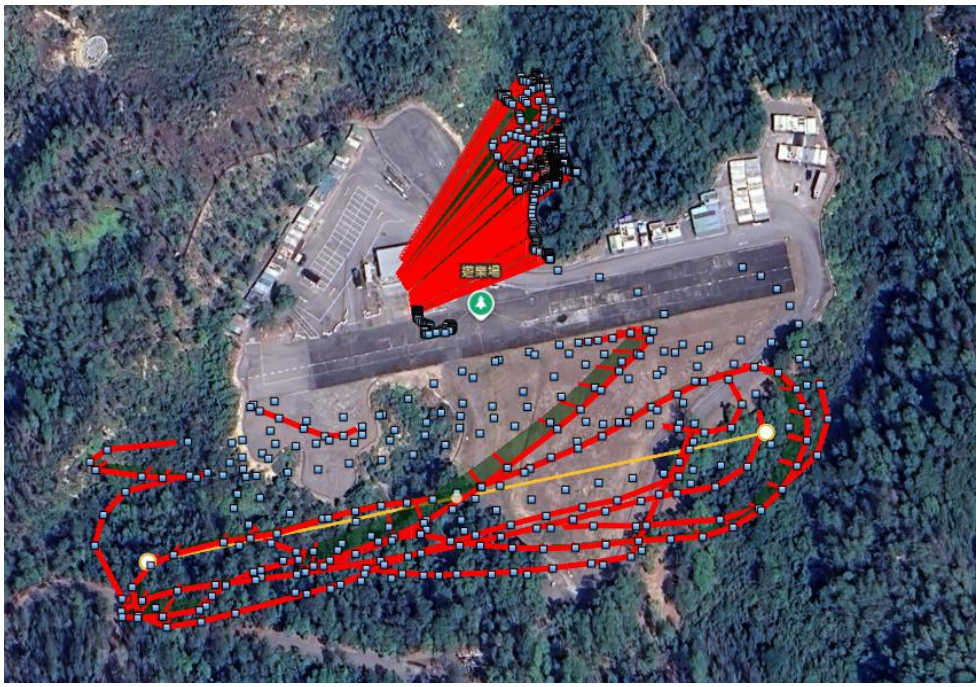


Figure 7.3 Flight route recorded by GPS

7.4 Pre-Flight Checklist

The HKUDBF team used the following checklist to ensure every flight test is safe and valuable. The checklist will continue be updated after each flight test when new findings in design requirements and challenges are discovered. The checklist involved Pre-flight Checklist, and a similar post-flight checklist would be included after the flight test ends. The team would check, verify and confirm the accuracy and thoroughness of the data collected from each flight test. Data will be stored on a shared cloud among the team for study to conduct design refinement for the aircraft.

Date:					
Power-off Preflight			**Power-on Preflight**		
Fuselage	Aircraft skin tear-free	Y/N	**Fuse out**	No fuse, Propulsion system off	Y/N
	Servos, linkages and horns secured	Y/N	**Fuse in, receiver switch off**	No response to command	Y/N
	Control surfaces stiffness	Y/N	**Range check**	Good signal quality at 30 steps away	Y/N
	Landing gears secured	Y/N	**Fail-safe**	Throttle closed	Y/N
Motors	Motor & firewall secured (check by wrench)	Y/N		Full up elevator	Y/N
Propeller	Prop undamaged	Y/N		Full right rudder	Y/N
	Prop direction (words to front)	Y/N		Full right aileron	Y/N
Batteries	Propulsion battery voltage		**Motor**	Rotating direction	Y/N
	Receiver battery charged?	Y/N	**Take off**		
	GPS battery charged	Y/N	**Round**		
	Remote controller charged	Y/N	**Data Collection**	Computer connected to GPS?	Y/N
Avionics	ESC/batteries/sensors secured	Y/N		All sensors' data on transmitter?	Y/N
	Servo moved correctly	Y/N		Cameraman ready	Y/N
	No loosen/uninsulated wires	Y/N		Start	
	All wires secured	Y/N		Number of Laps	
M1	No payload	Y/N		End	
	Take off weight			Duration	
M2	Passengers and cargoes placed?	Y/N	**Post Flight**		
M3	Parts' position	Y/N	**Disarm**	Transmitter throttle cut	Y/N
	Clamp locked?	Y/N		Propulsion battery disarmed	Y/N
Wing tip test	Wing strength	Y/N			
	Cg correct	Y/N	**Data Collection**	Battery Voltage	=
	No of weight added (+: nose;-: tail)			GPS disconnected?	Y/N
Take off weight			**Visual Inspection**	Damage observed	Y/N

Table 7.4 Pre-flight Checklist

8 Performance Results

8.1 Subsystem Performance

8.1.1 Propulsion

The results of the propulsion test show that the EOLO13x7, being a high efficiency propeller, generated 1.72lbs more thrust than Sail 13x7 (wooden propeller). Therefore, the team used the EOLO propeller. Moreover, when the pilot simulated the flight test and ran the propulsion system for 5 minutes by using the EOLO 12x6.5 propeller, the actual energy consumption (61.90Wh) was found to be close to the specification's the energy consumption (63.43Wh) at 80% of the maximum current at the battery's nominal voltage (22.2V). Based on the above, battery capacity selection was refined to 4500mAh

		Experimental		Theoretical	
Propeller	**Battery**	**Maximum Thrust (lbs)**	**I_max (A)**	**Maximum Thrust (lbs)**	**I_max (A)**
Sail 13x7	ATECH Infinity 3300mAh 6s battery 55C	7.92	47.56	11.11	65
EOLO 13x7	GAONENG 3300mAh 6s battery 100C	9.64	52.77	11.11	65
EOLO 12x6.5	GAONENG 3300mAh 6s battery 100C	8.18	43.16	9.64	50.5

Table 8.1.1 Propulsion test result

8.2 Complete Aircraft Flight Performance

The following table compares the complete performance of the two prototypes of Griffin:

Prototype 1 (48 passengers +16 cargoes;13x2.5ft banner)				
Properties	M1	M2	M3	GM
Payload (lbs)	N/A	0.509	N/A	N/A
Banner (lbs)	N/A	N/A	0.551	N/A
Take-off Weight (lbs)	8.17	26.27	8.53	N/A
Cruise Speed (ft/s)	83.66	N/A	59.06	N/A
Number of Laps	3	N/A	5	N/A
Mission Time (sec)	86.47	N/A	293	505
Prototype 2 (3 passengers + 1 cargo;13x2.5ft banner)				
Properties	M1	M2	M3	GM
Payload (lbs)	N/A	0.509	N/A	N/A
Banner (lbs)	N/A	N/A	0.551	N/A
Take-off Weight (lbs)	5.29	5.8	5.95	N/A
Cruise Speed (ft/s)	105.0	105.0	59.06	N/A
Number of Laps	3	9	5	N/A
Mission Time (sec)	93.75	281.25	293	25

Table 8.2 Complete Aircraft Flight Performance Comparison

For prototype 1 as shown in figure 8.2a, the mission 2 performance could not be recorded as prototype 1 was lost due to sudden damage of the electronic equipment on the aircraft during the flight test of examining the aircraft's performance for mission 3 as shown in figure 8.2b.





Figure 8.2a and 8.2b

For prototype 2, it completed the number of laps as predicted during the preliminary design stage. Both prototypes were tested at an altitude of around 65ft to 70ft. The maximum g-force the prototype withstood during the flight test was 6g. The team would continue focusing on improving the aerodynamic performance and mission performance for the final competition aircraft and conduct regular practices of ground mission before the flyoff day.

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